General Chemistry I

Chemistry: The central science

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Lectures 11,12

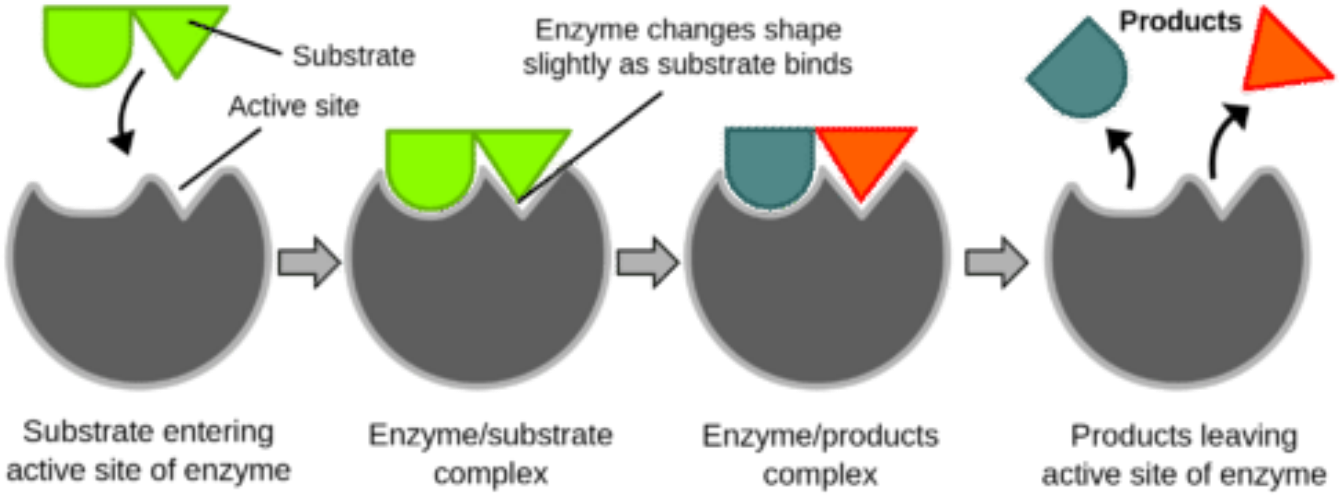
Bonding theories and **Molecular geometry**

Baby Puzzles



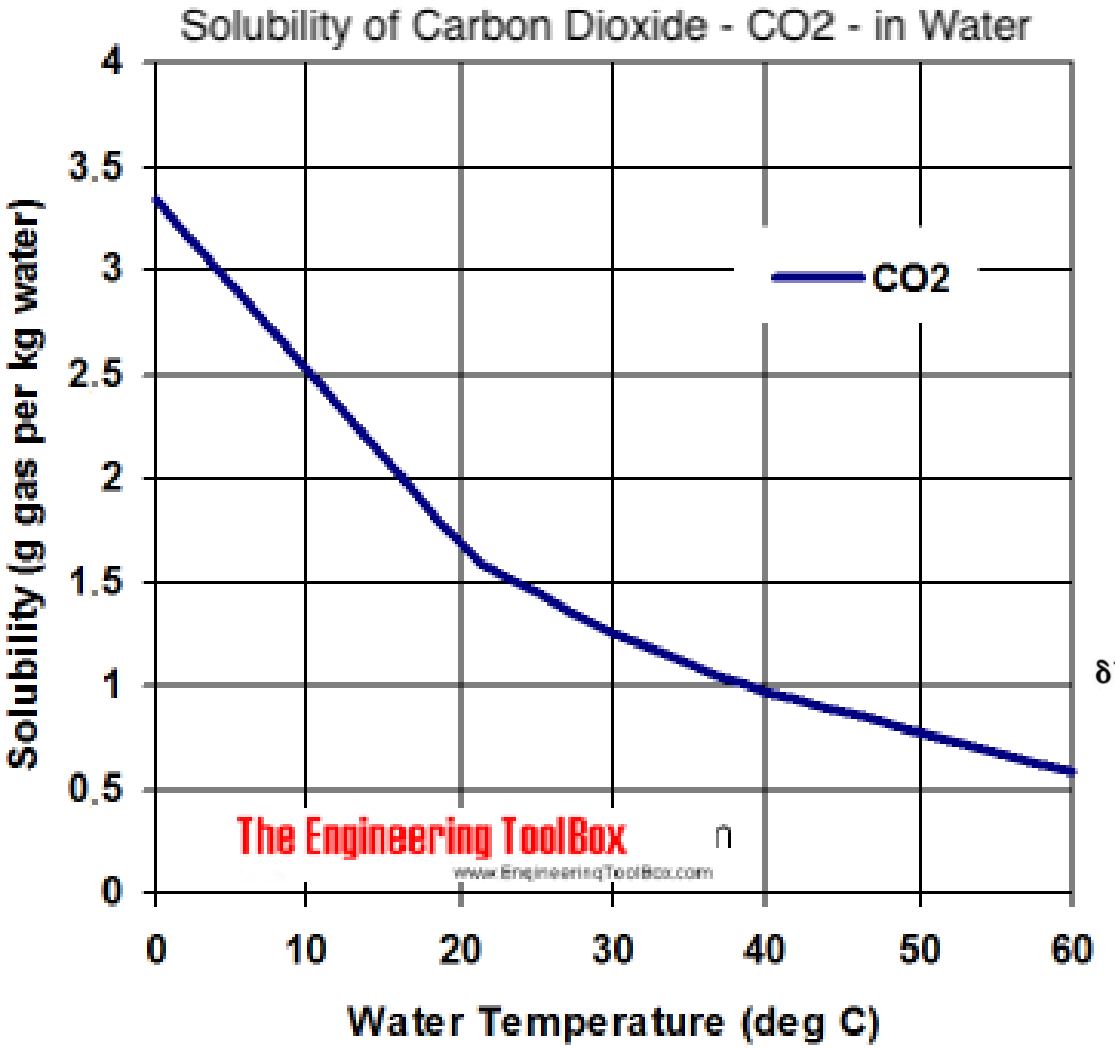
Why is molecular shape important?

Enzyme – Biological Catalyst



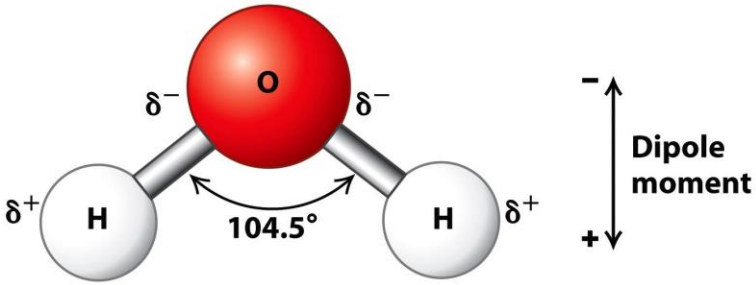
https://chem.libretexts.org/Courses/Georgia_Southern_University/CHEM_1152%3A_Survey_of_Chemistry_II_%28GSU_-_Dr._Osborne%29/08%3A_Proteins/8.05%3A_Enzymes_-_Biological_Catalysts

Why is molecular shape important?





Carbon dioxide

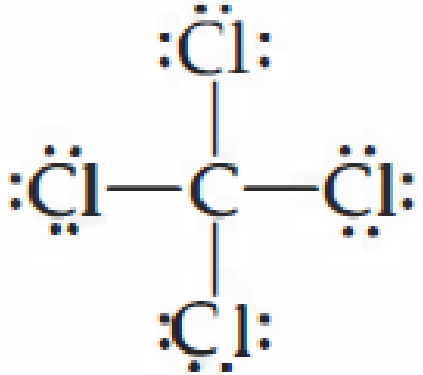


Content

1. Molecular shapes
2. VSEPR Model
3. Molecular shape and molecular polarity
4. Covalent bonding and orbital overlap
5. Hybrid orbitals
6. Multiple bonds
7. Molecular orbitals

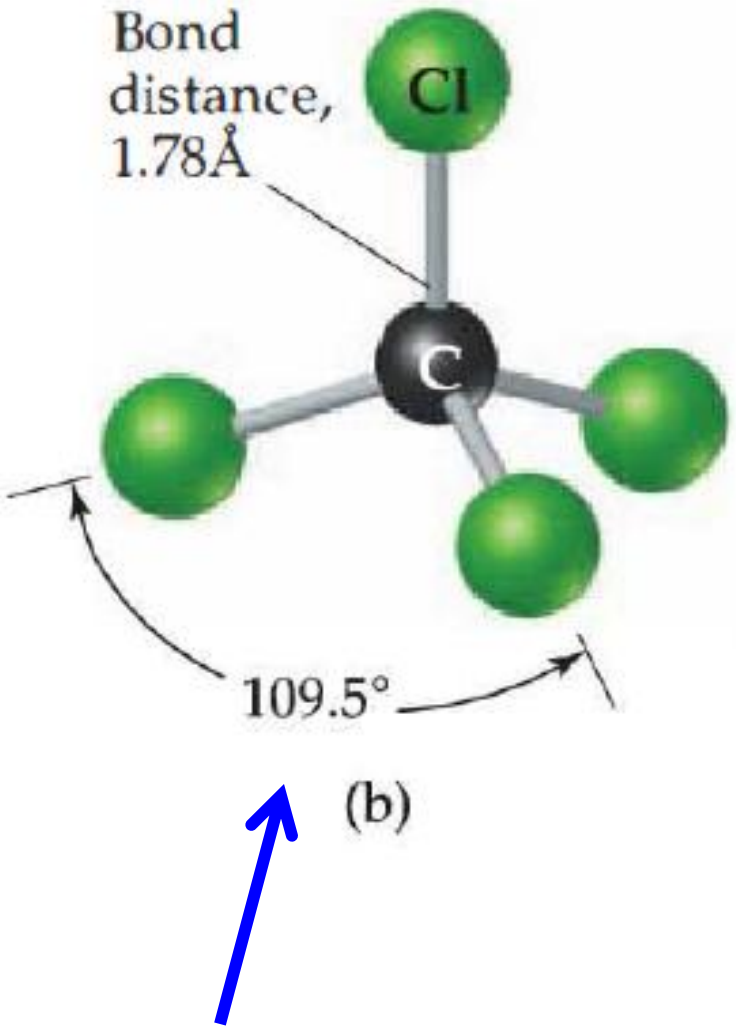
1. Molecular shapes

Lewis structure





Tetrahedron



Bond angle

Molecular shapes of AB_n

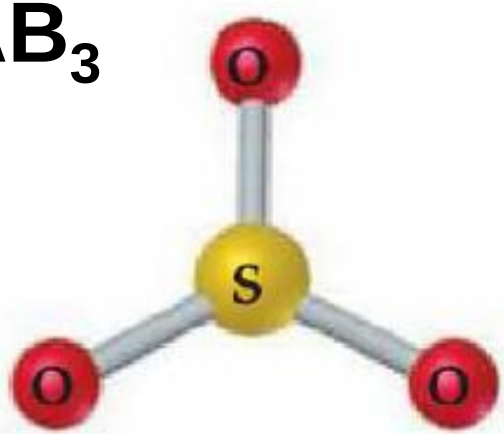
AB_2



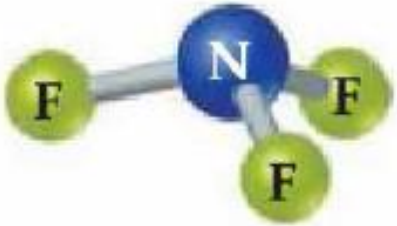
Linear

Bent

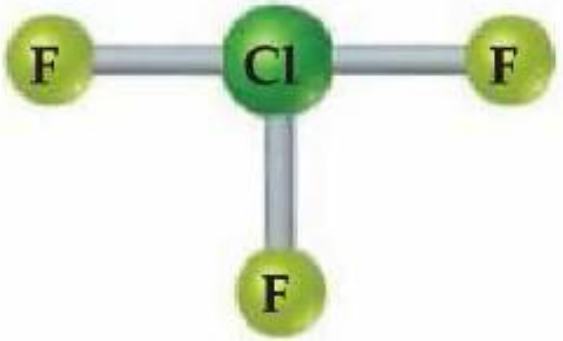
AB_3



Trigonal planar

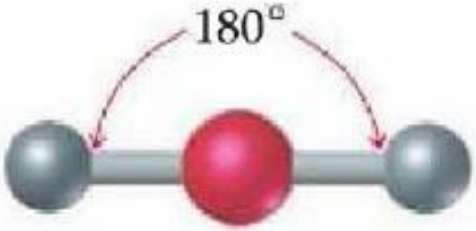


Trigonal pyramidal

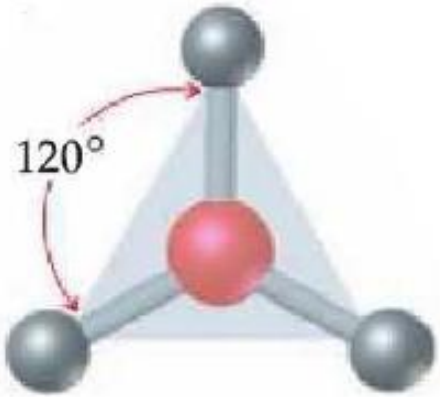


T-shaped

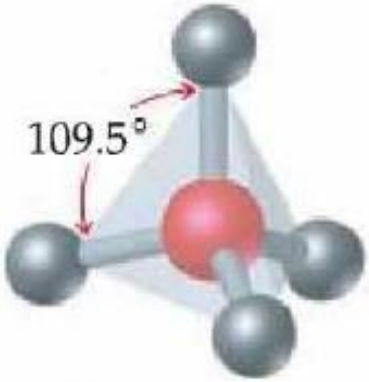
Molecular shapes of AB_n



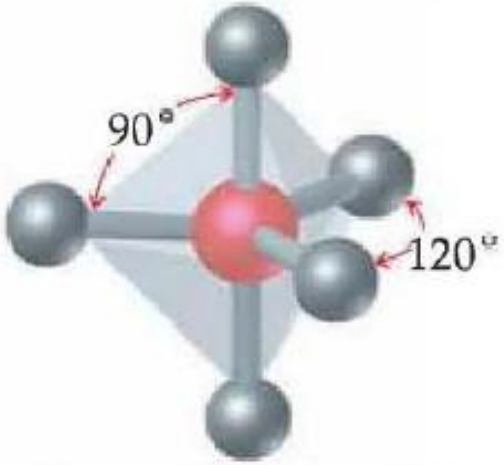
Linear



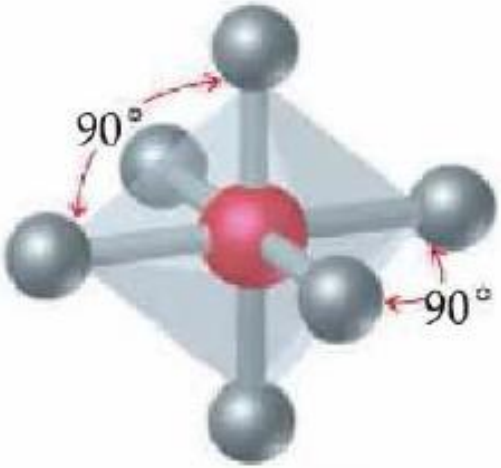
Trigonal planar



Tetrahedral



Trigonal bipyramidal

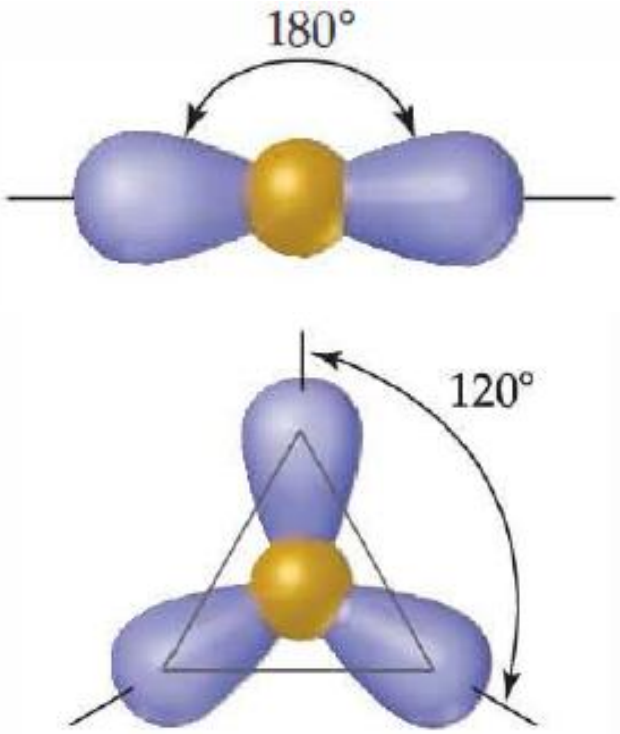


Octahedral

Metal ion in water: $[M(H_2O)_6]^{n+}$

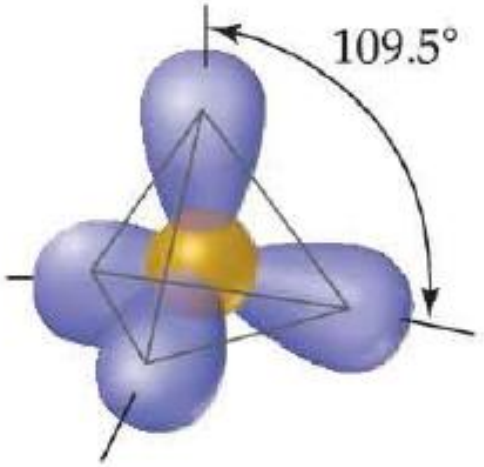
Valence-Shell Electron-Pair Repulsion (VSEPR) model

Electron domains repel one another



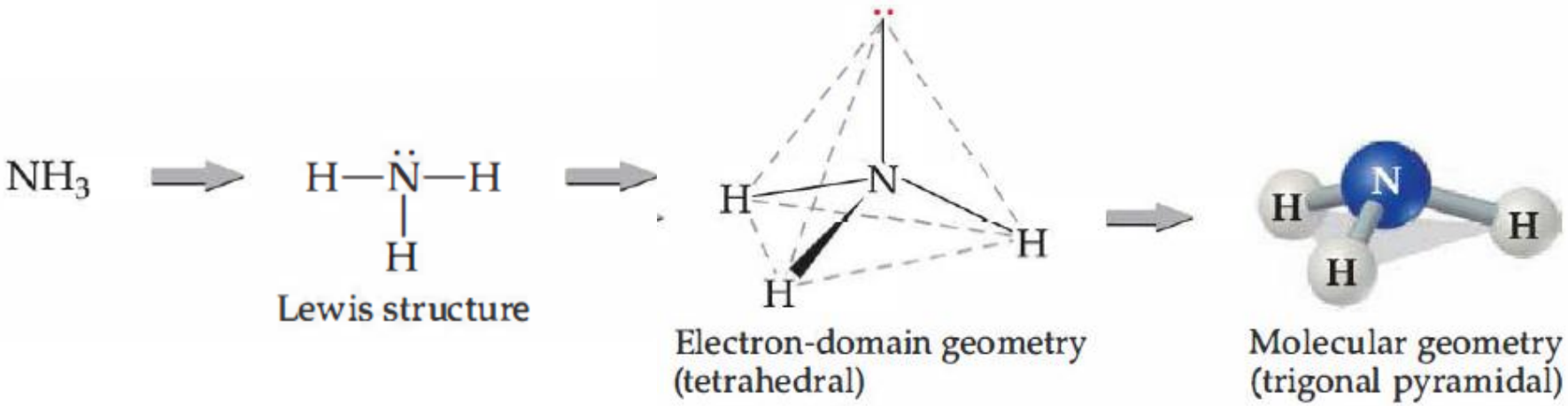
Linear

Trigonal planar



The best arrangement of a given number of electron domains is the one that minimizes the repulsion among them

Using VSEPR model to predict the shapes of molecules or ions



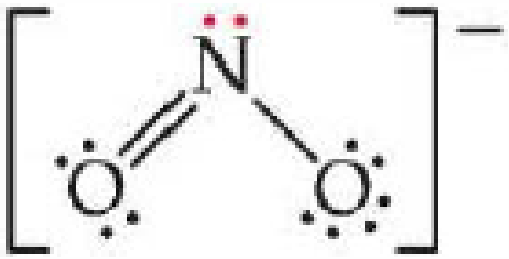
1. Draw Lewis structure

2. Determine electron-domain geometry

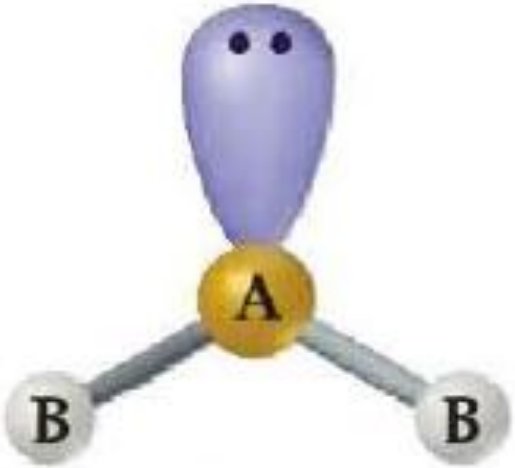
3. Determine molecular geometry

Using VSEPR model to predict the shapes of molecules or ions

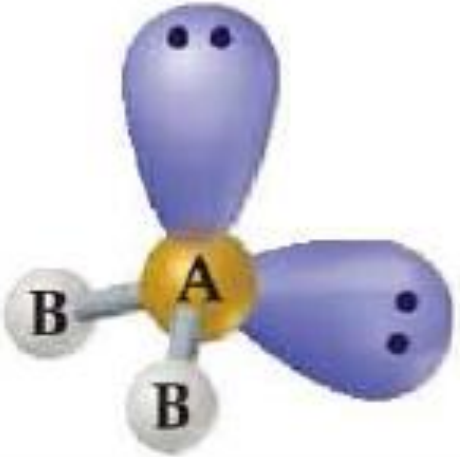
Bond angles deviate from the ideal angles when the surrounding atoms and electron domains are not identical







Bent



Bent

Example

Use the VSEPR model to predict the molecular geometry of O_3

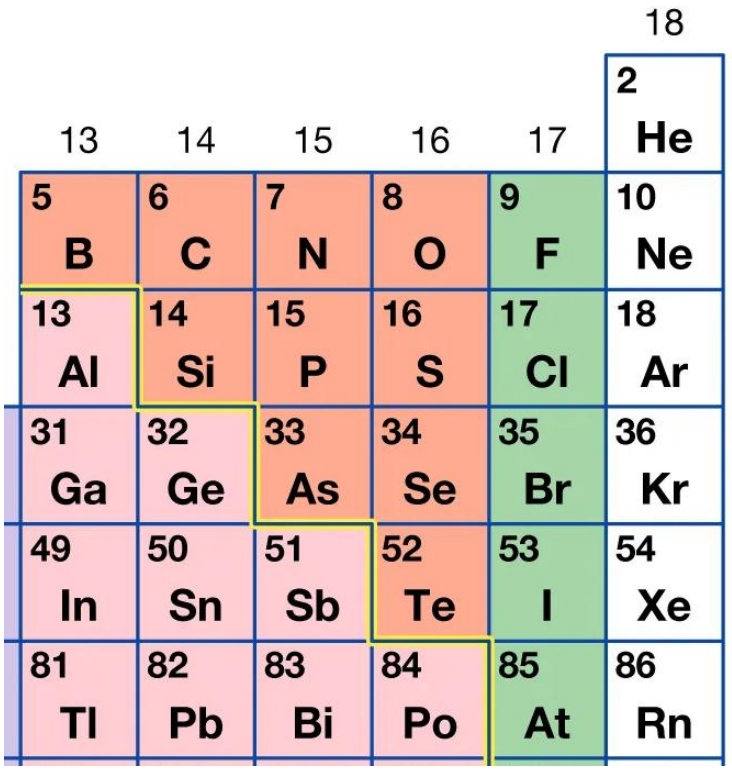
Geometry (shape) of $SnCl_3^-$ anion is:

A) Tetrahedral

B) Trigonal planar

C) Trigonal pyramidal

D) Square planar



Which of following statement(s) is (are) correct for the NH_3 molecule?

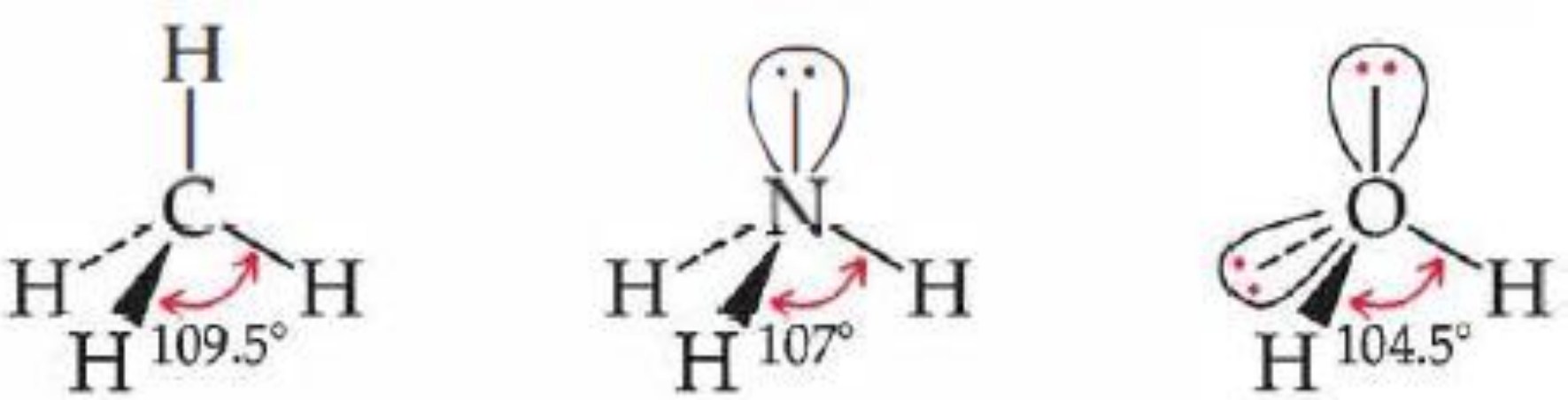
A) It adopts trigonal pyramidal shape

B) It is polar compound

C) It acts as a Lewis acid

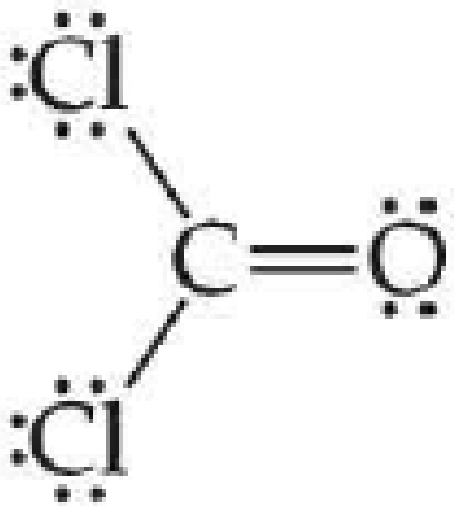
D) It is not soluble in water

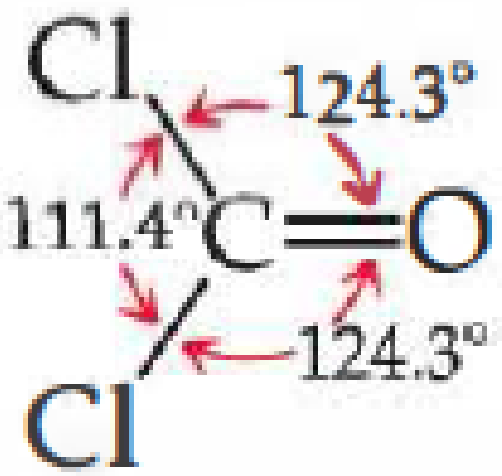
Effect of Nonbonding Electrons on Bond Angles



Non bonding electron pairs is spread out and exert greater repulsive forces on adjacent electron domains

Effect of Multiple Bond on Bond Angles

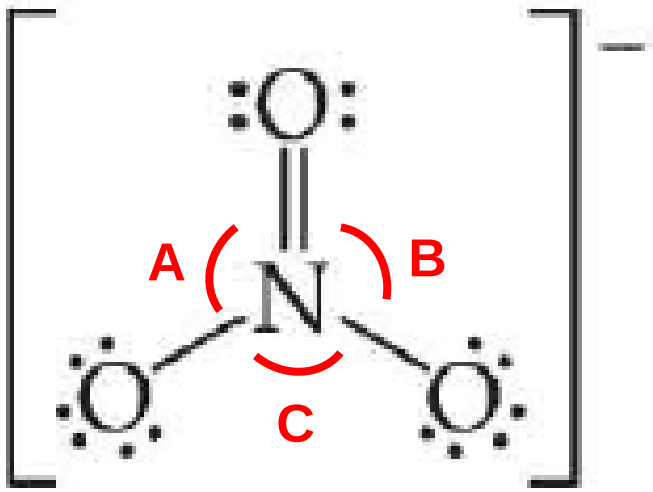




Electron domains in multiple bonds exert a greater repulsive force than single bonds

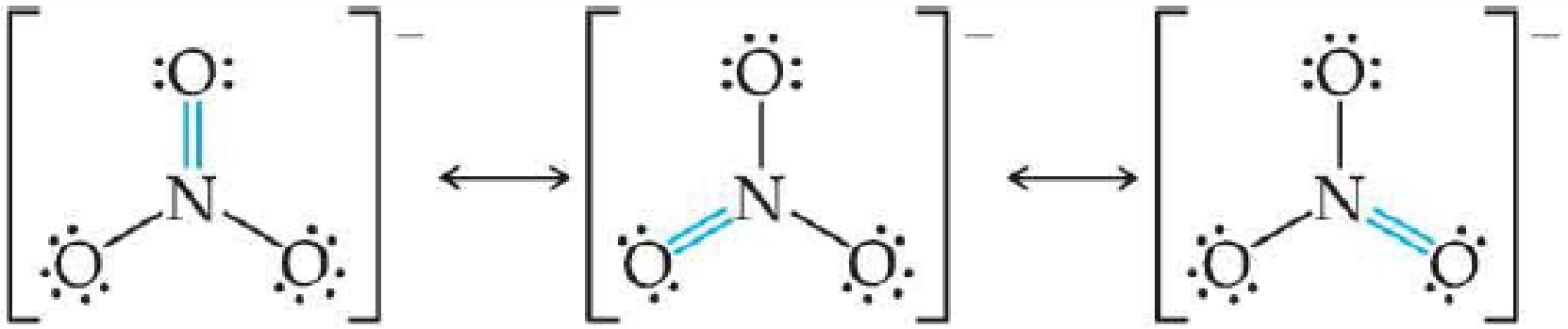
Special case

Nitrate ion



Which angle would has largest value?

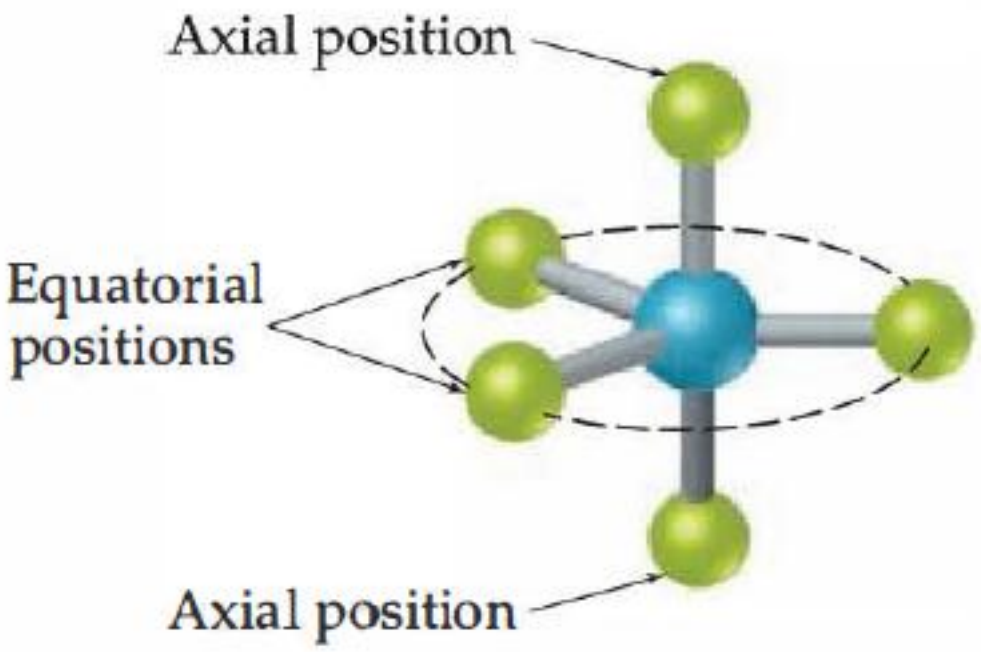
Resonance structures



Resonance structures are used when two or more equally valid Lewis structures can be written

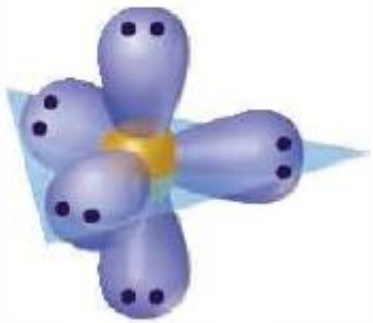
VSEPR Model of Expanded Valence Shells

Central atom with more than 8 valence electrons

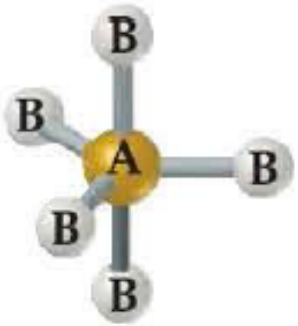


If a molecule has nonbonding electron domains, they will occupy the equatorial positions.

Molecular Shapes for Molecules with Five and Six Electron Domains around the Central Atom

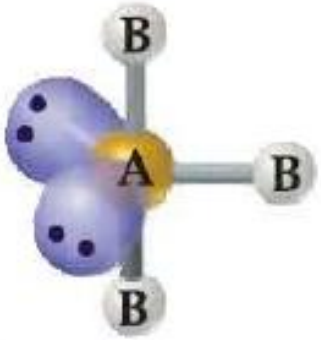


Trigonal bipyramidal



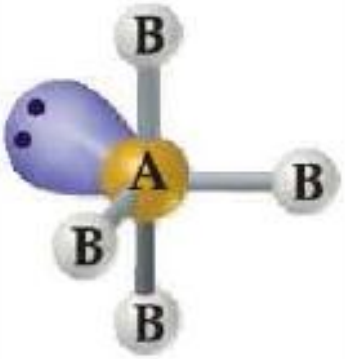
Trigonal bipyramidal

PCl_5



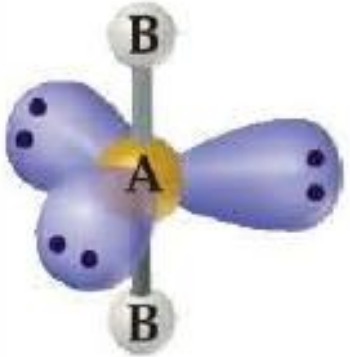
T-shaped

ClF_3



Seesaw

SF_4



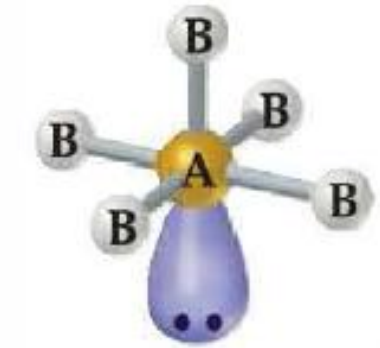
Linear

XeF_2

Molecular Shapes for Molecules with Five and Six Electron Domains around the Central Atom



Octahedral



Square pyramidal

BrF_5

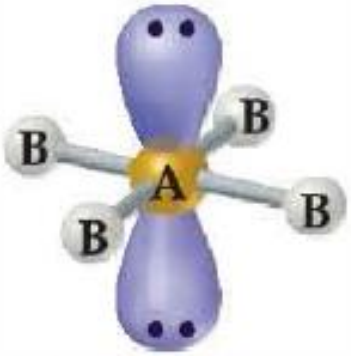


Octahedron



Octahedral

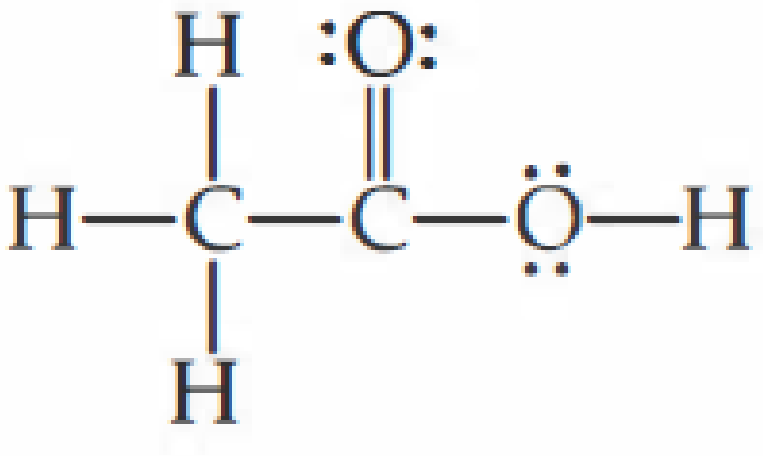
SF_6



Square planar

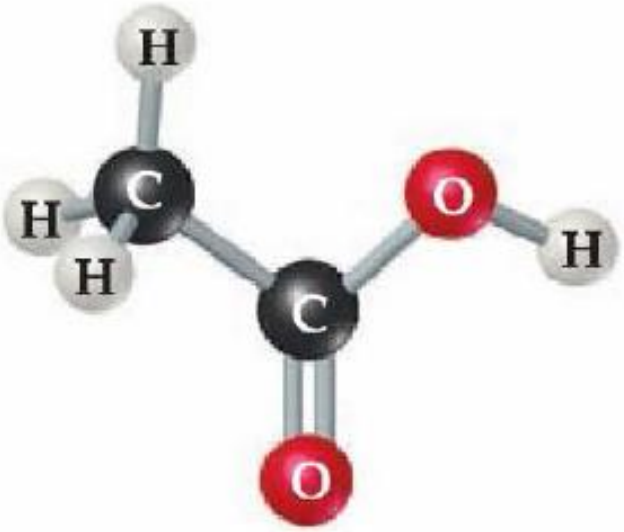
XeF_4

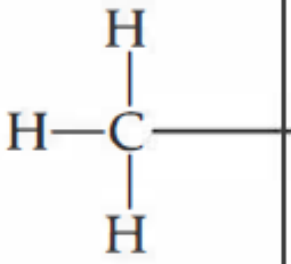
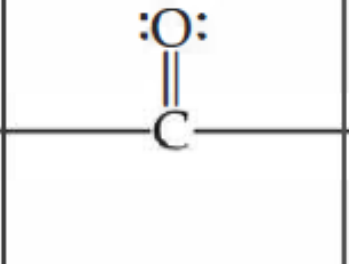
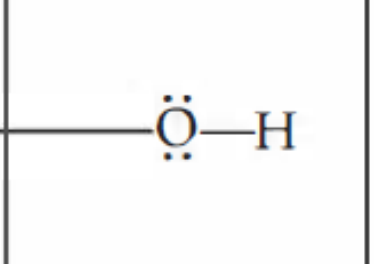
Shapes of Larger Molecules



1. Find interior atoms: C, C, O
2. Predict geometry about each of these atoms individually

Shapes of Larger Molecules

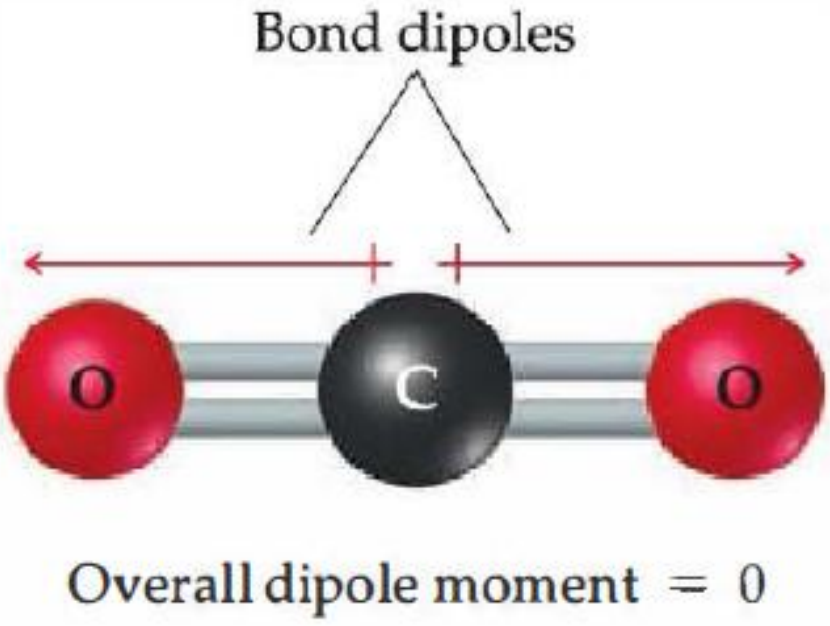


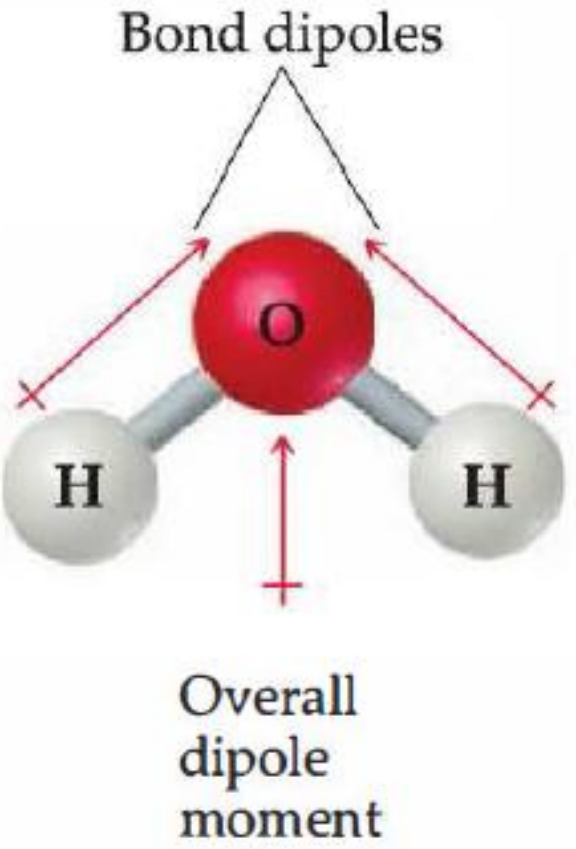
			
Number of electron domains	4	3	4
Electron-domain geometry	Tetrahedral	Trigonal planar	Tetrahedral
Predicted bond angles	109.5°	120°	109.5°

Molecular Shape and Molecular Polarity

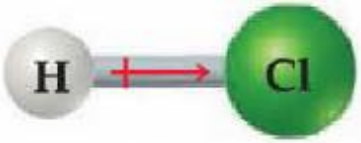
Dipole moment depends on both the polarities of the individual bonds and the geometry of the molecule

Bond dipole: dipole moment due only to the two atoms in that bond

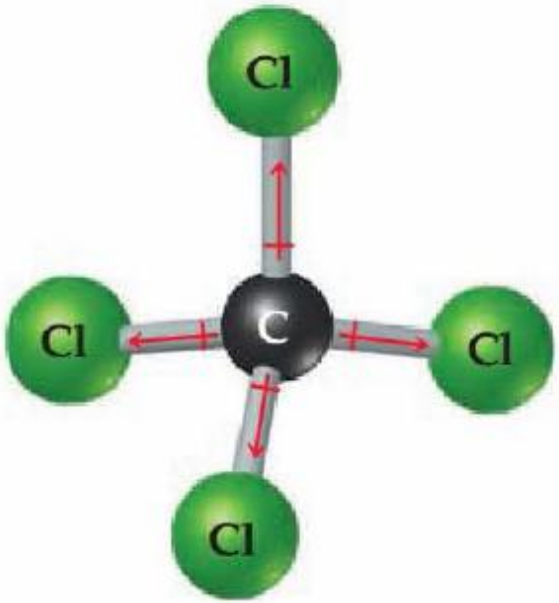




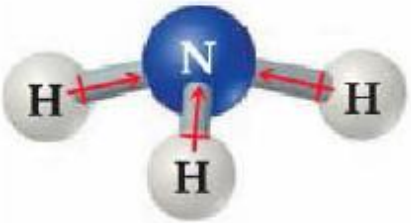
Which of these molecules have polar bonds?



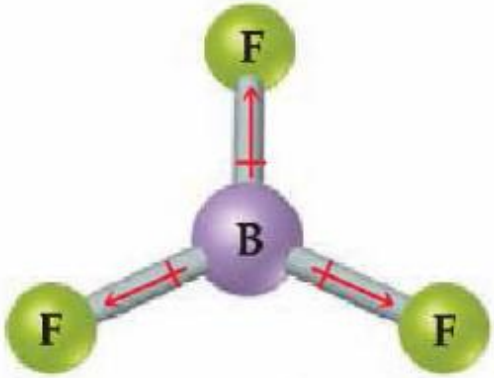
Polar



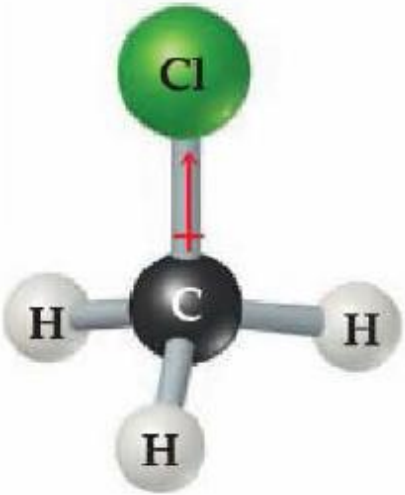
Nonpolar



Polar



Nonpolar



Polar

Which of following molecules is (are) polar:

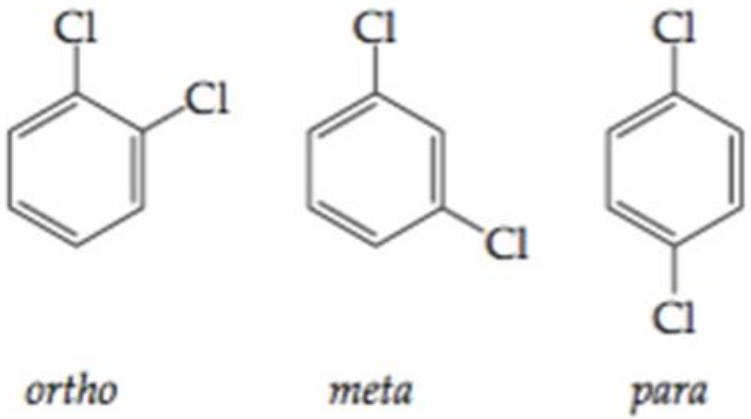
A) IF

B) CS_2

C) PCl_3

D) SF_6

Dichlorobenzene, $C_6H_4Cl_2$, exists in three forms (isomers) called *ortho, meta,* and *para*. Which of these has a zero dipole moment?



A) Ortho

B) Metal

C) Para

D) All of these isomers

Which of following compounds would have largest H-X-H bond angle (X = C, N or O)?

A) CH_4

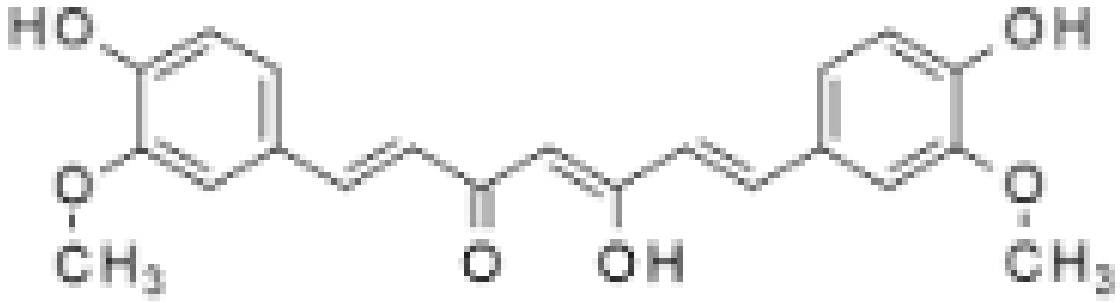
B) NH_3

C) H_2O

Curcumin



Make it soluble in WATER????



(1*E*,6*E*)-1,7-Bis(4-hydroxy-3-methoxyphenyl)-1,6-heptadiene-3,5-dione

PX_5 chemistry

1. What would be geometry of PF_5 and PCl_5 molecules?

2. Which of these two compounds (PF_5 and PCl_5) is more stable?

3. In the solid phase, PCl_5 transforms into ionic compound which is more stable.

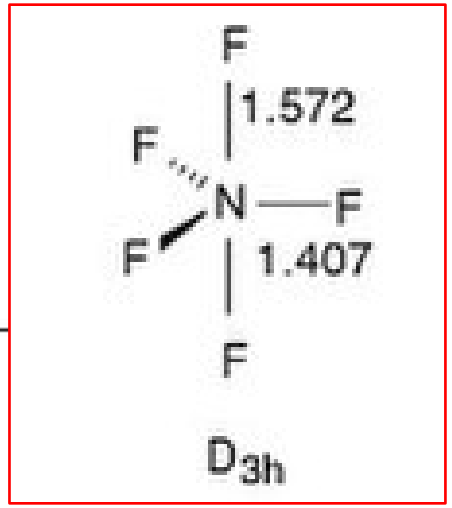
Predict the structure and geometry of the ionic compound product.

AIP | The Journal of Chemical Physics

The electronic structure and stability of NF5 and PF5

H. H. Michels and J. A. Montgomery Jr.

Citation: The Journal of Chemical Physics **93**, 1805 (1990); doi: 10.1063/1.459108
View online: http://dx.doi.org/10.1063/1.459108
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Articles you may be interested in

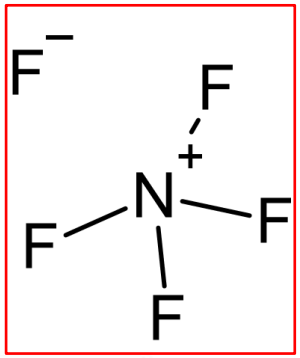
Negative ion reactions with PF5 and the electron affinity of PF5
J. Chem. Phys. **100**, 7200 (1994); 10.1063/1.466918

Bonding and ionization energies of N–F and P–F compounds
J. Chem. Phys. **81**, 6166 (1984); 10.1063/1.447571

Infrared Spectrum of PF5
J. Chem. Phys. **24**, 920 (1956); 10.1063/1.1742656

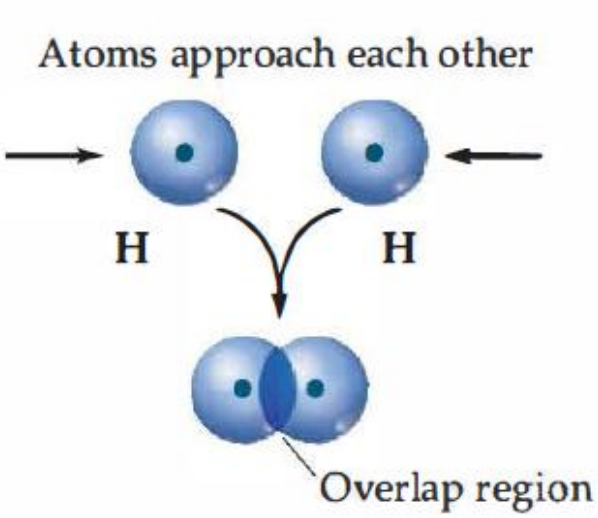
Erratum: The Infrared Spectra of NF3 and PF3
J. Chem. Phys. **21**, 1426 (1953); 10.1063/1.1699269

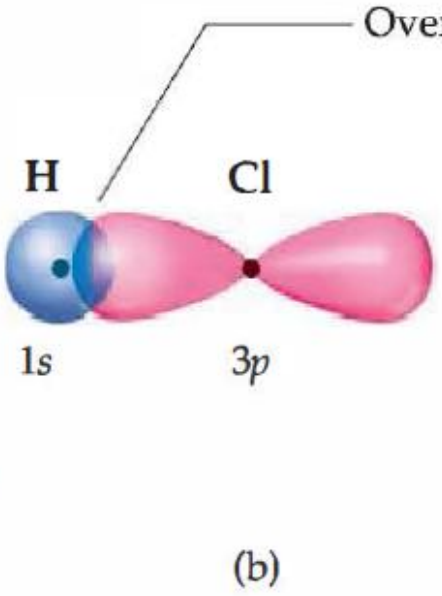
The Infrared Spectra of NF3 and PF3
J. Chem. Phys. **20**, 1716 (1952); 10.1063/1.1700275

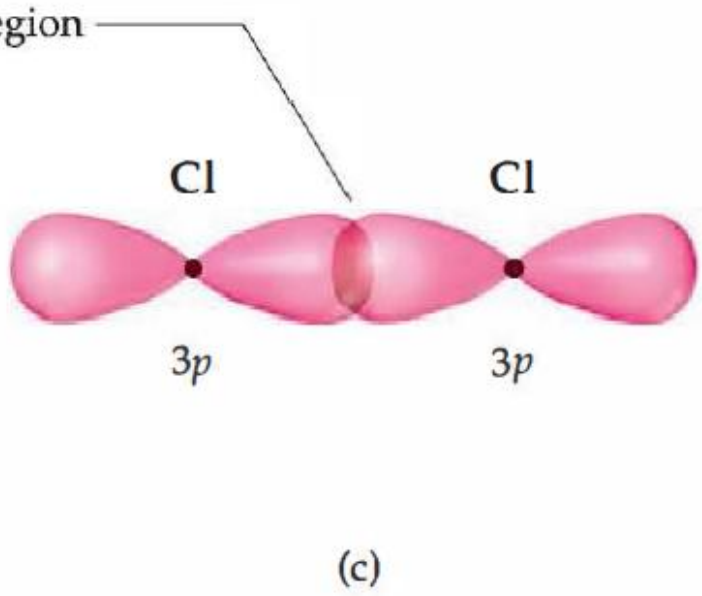


Covalent Bonding and Orbital Overlap

Valence bond theory

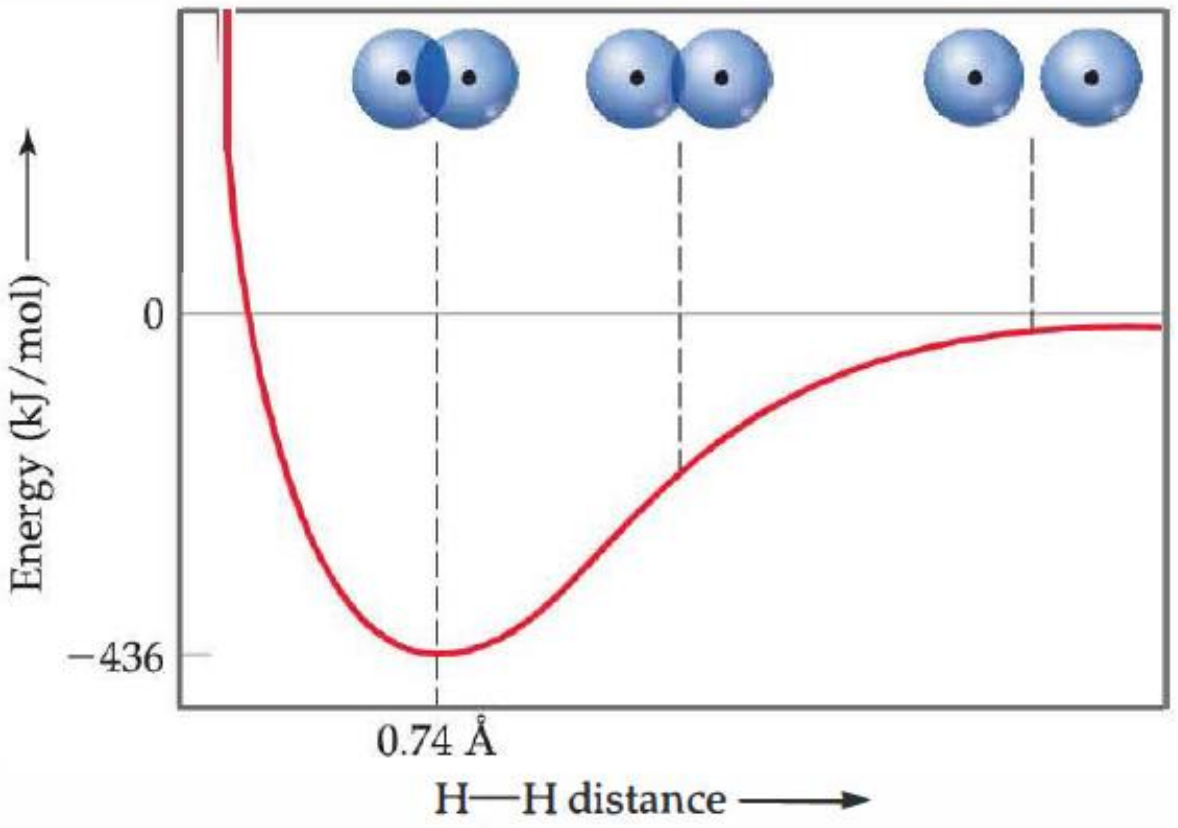




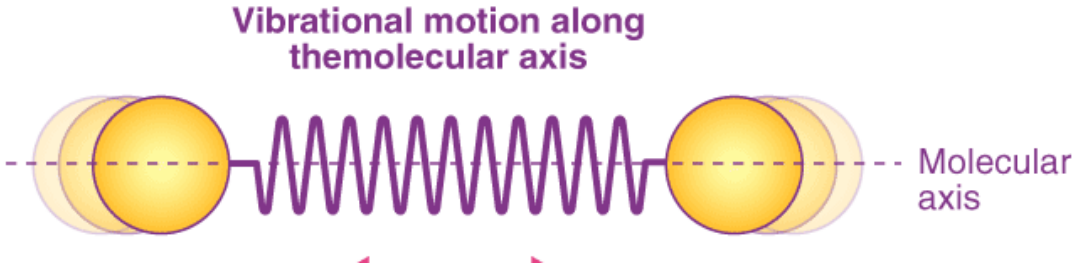


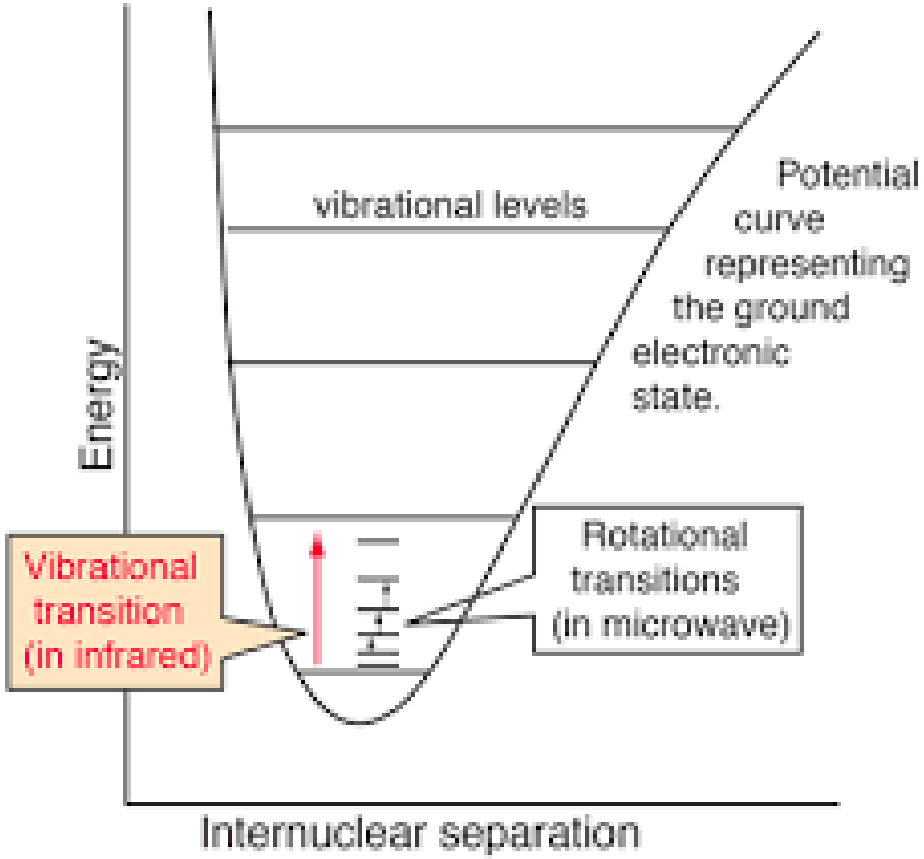
Optimum bond length in covalent bond

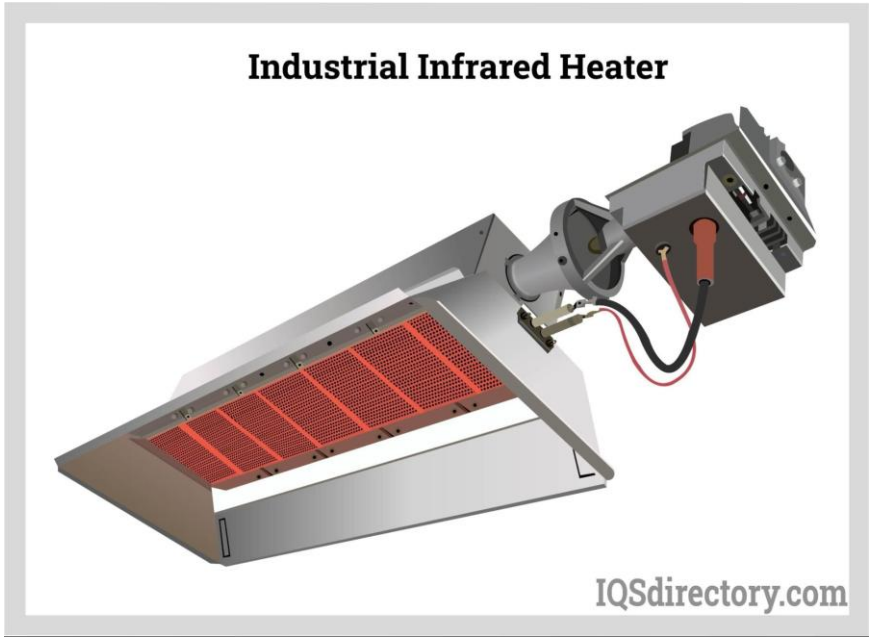
The bond length in covalent bond is the distance at which attractive forces between unlike charges are balanced by repulsive forces between like charges



Optimum bond length in covalent bond



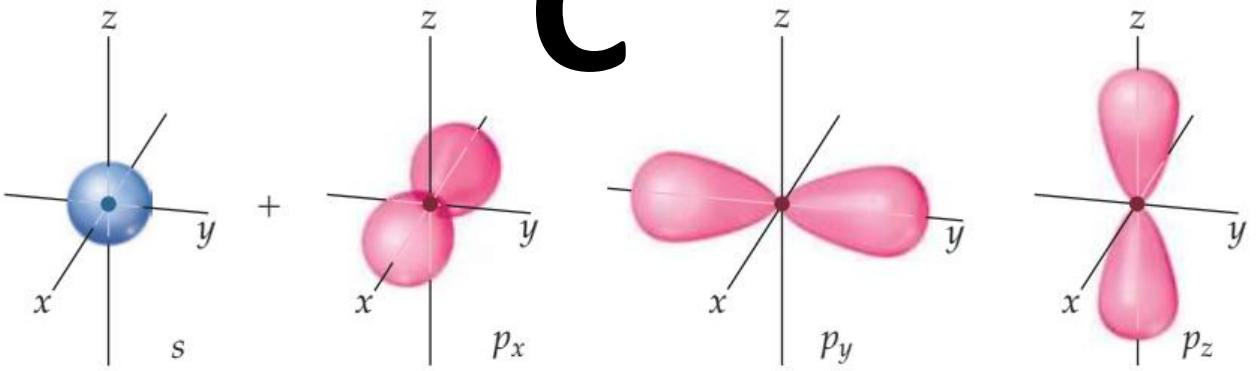




Hybrid Orbitals

$C + 4H \rightarrow CH_4$

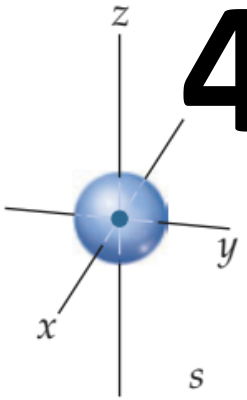
C



One *s* orbital

Three *p* orbitals

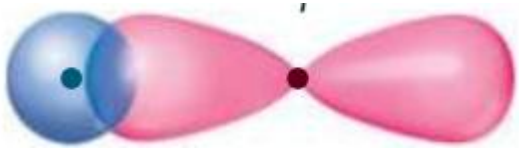
4H



One *s* orbital

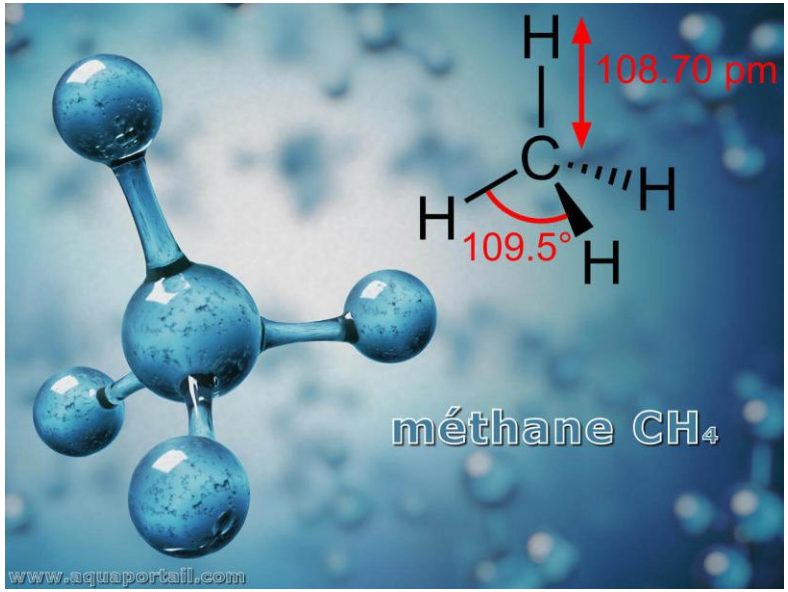


1 s - s

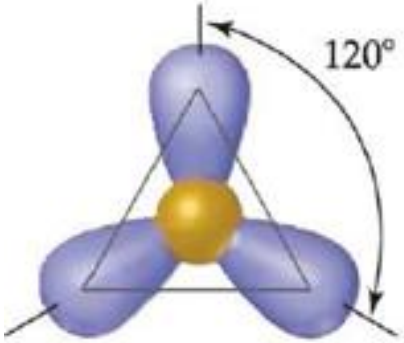


3 s - p

≠

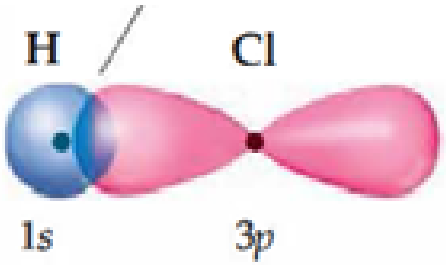


Valence-shell electron-pair repulsion (VSEPR) model



- Based on electron domain repulsion force
- Neglect atomic orbitals
- Can predict molecular geometries

Valence bond theory



- Based on atomic orbitals' shapes
- Cannot explain **experimental observations**

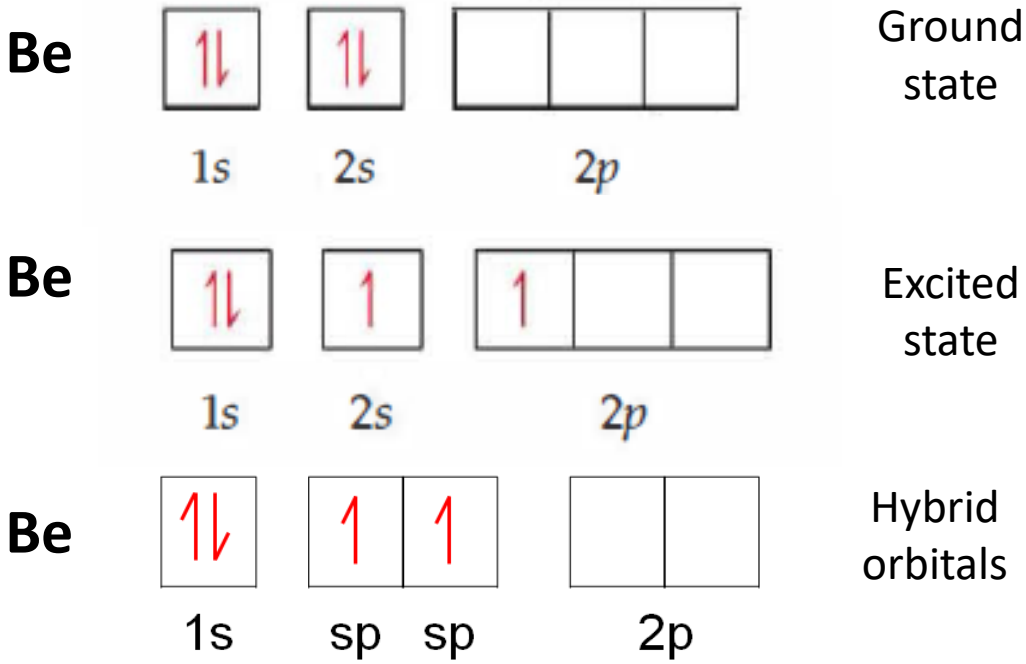
*To resolve the inconsistencies, a **new concept** was introduced*

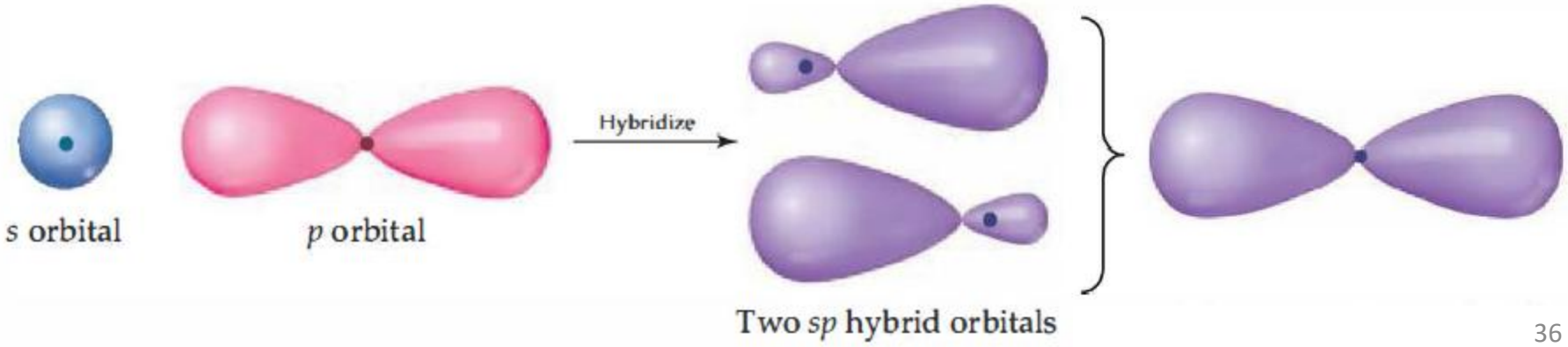
Hybrid orbitals

- Explains the observed orbitals in covalent bonding
- Works well for Carbon compound
- Use frequently in organic chemistry



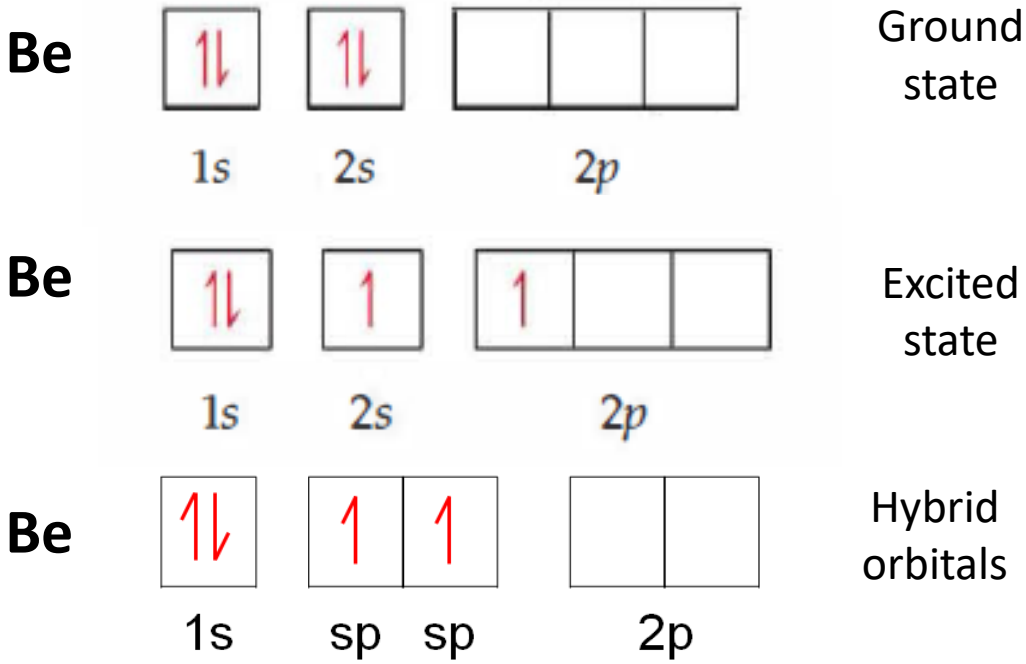
sp Hybrid Orbitals

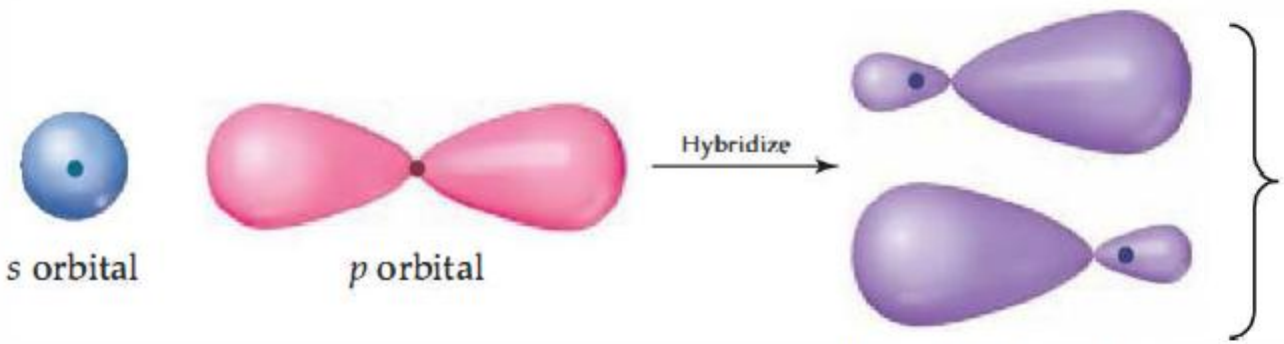






sp Hybrid Orbitals



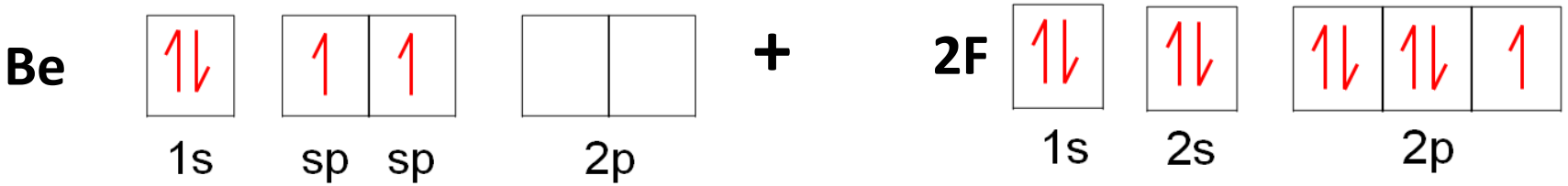


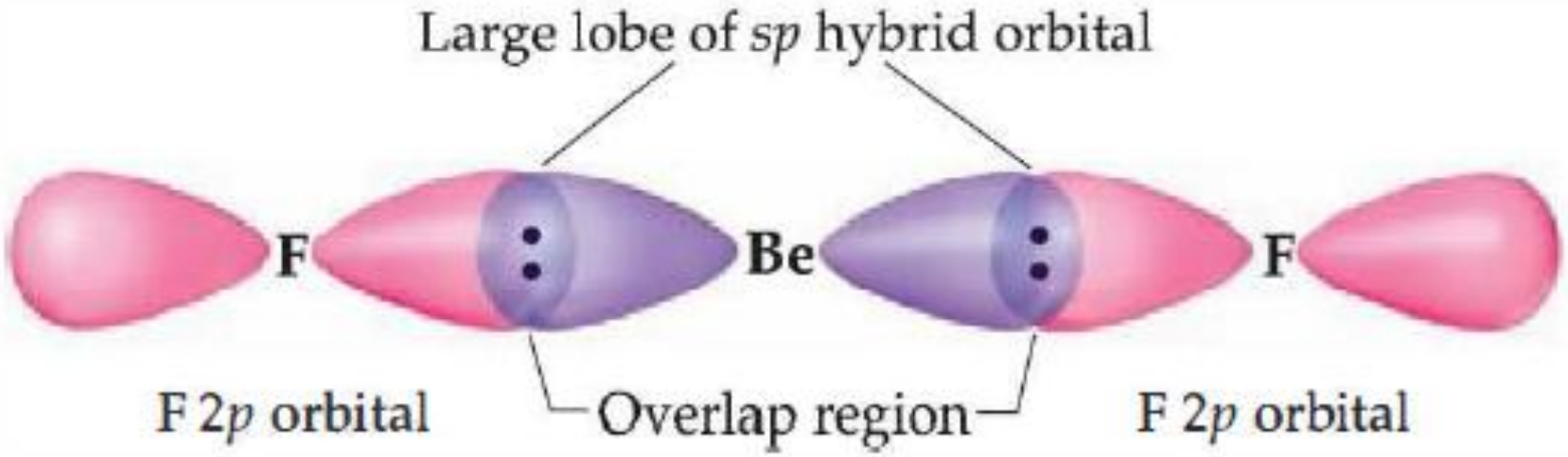
(50% s and 50% p characters)



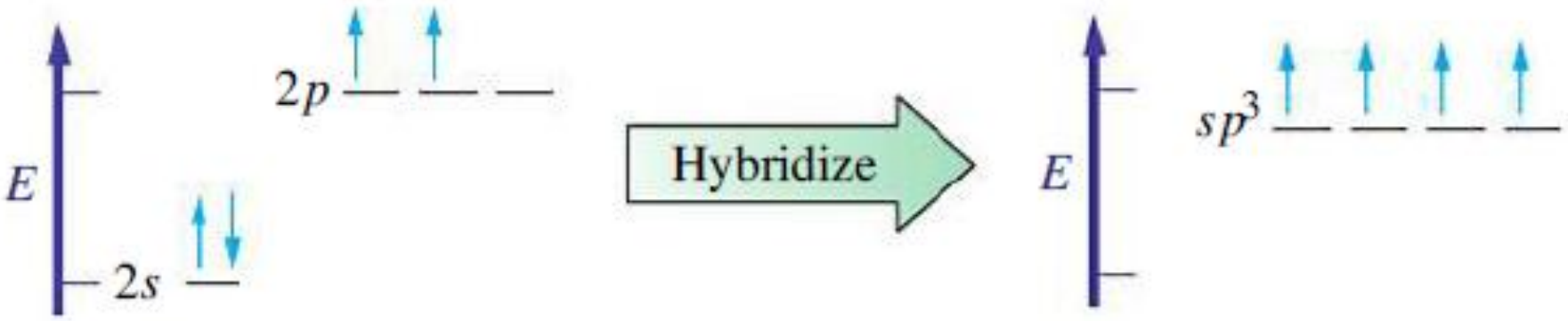


sp Hybrid Orbitals





Hybrid Orbitals



sp hybrid orbitals = 1s orbital + 1p orbital

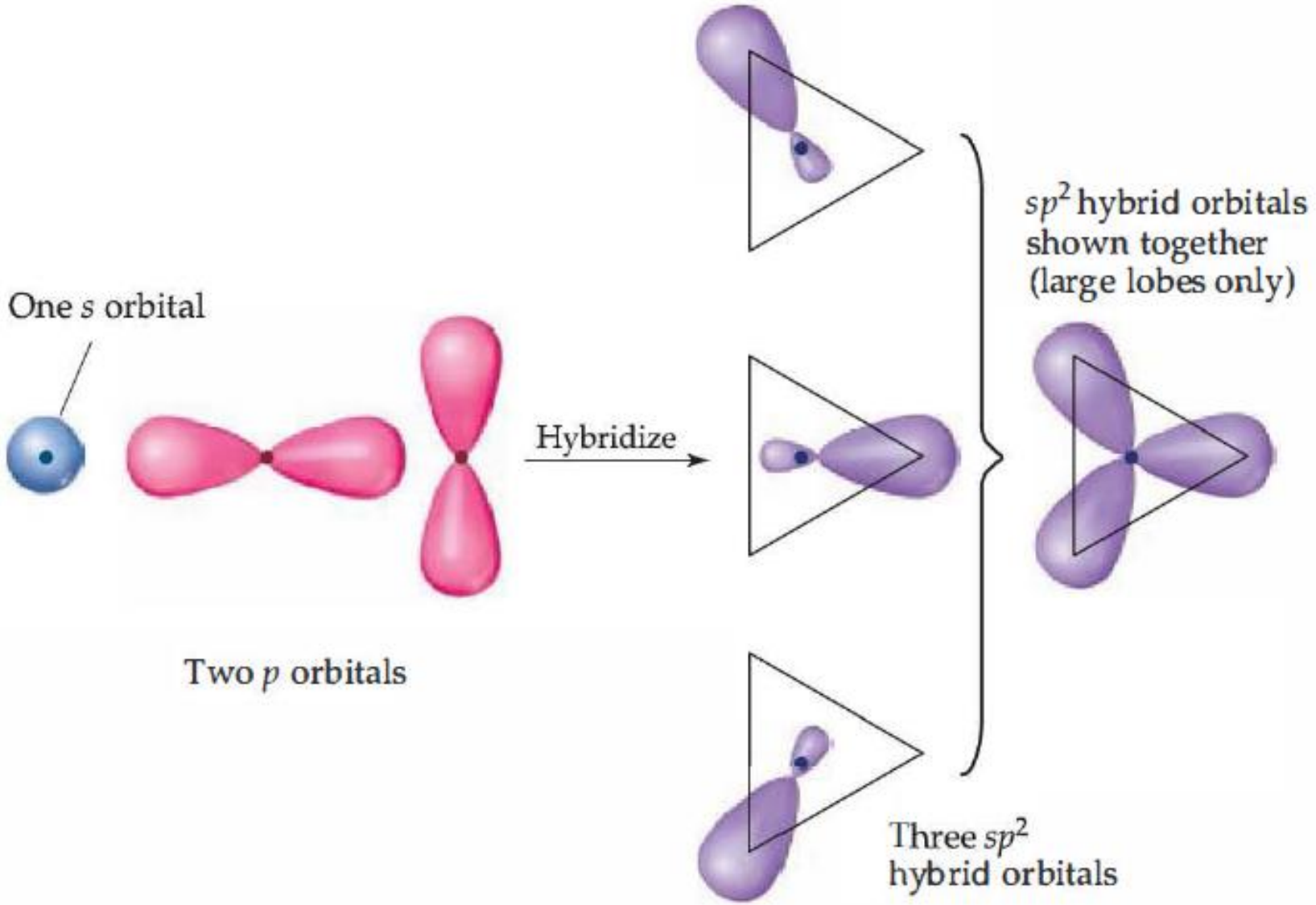
sp^2 hybrid orbitals = 1s orbital + 2p orbitals

sp^3 hybrid orbitals = 1s orbital + 3p orbitals

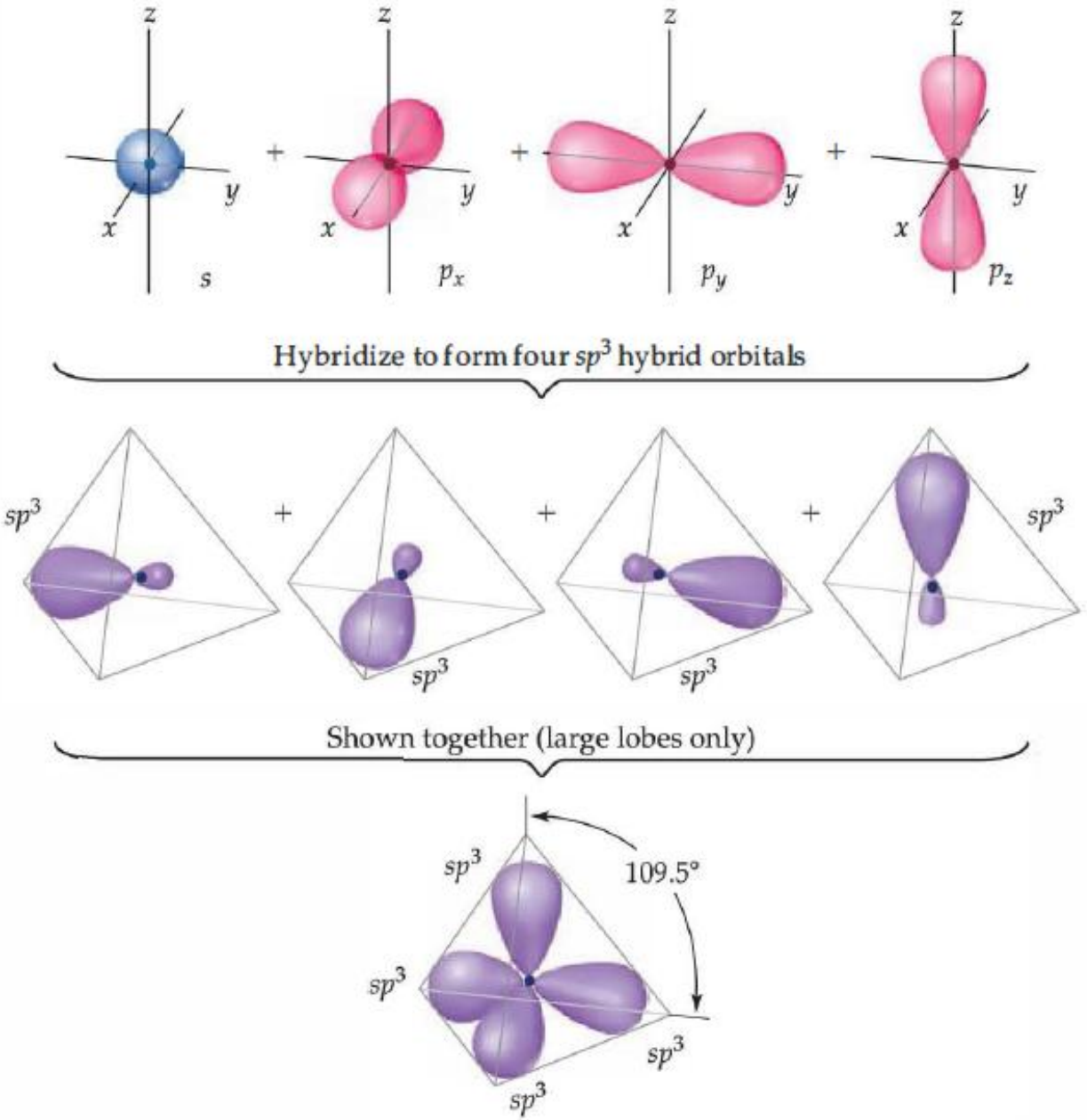
Remark:

In term of energy: sp < sp^2 < sp^3

sp^2 Hybrid Orbitals

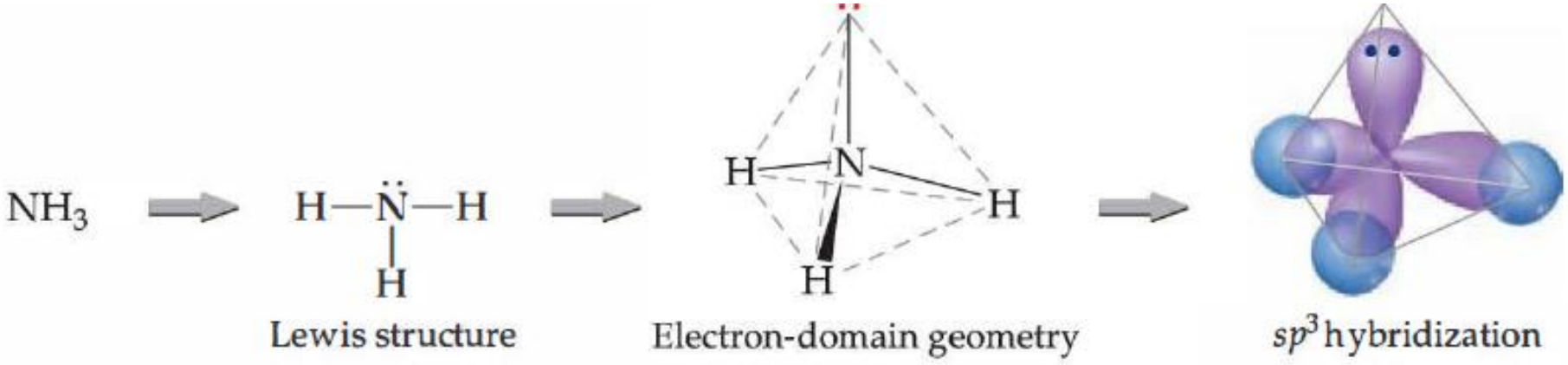


sp^3 Hybrid Orbitals

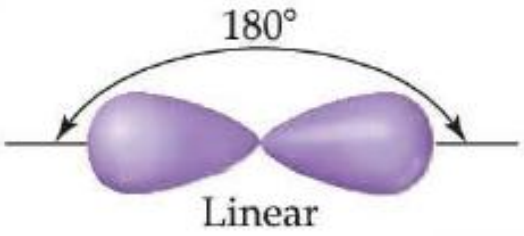
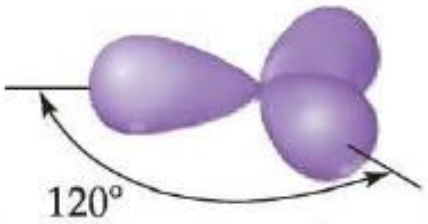
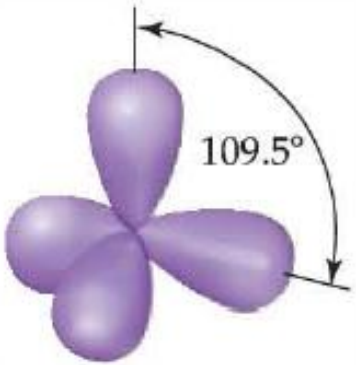


Predict Hybrid Orbitals

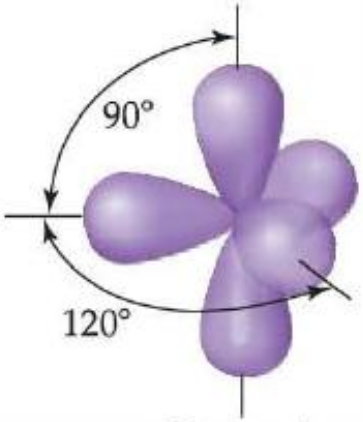
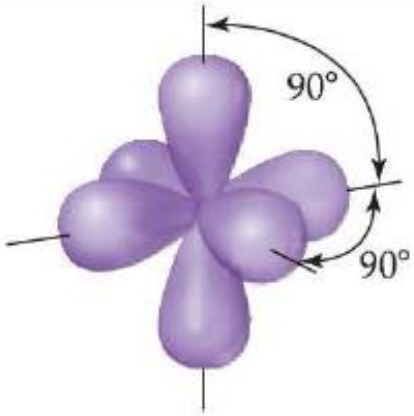
1. Draw the Lewis structure for the molecule or ion
2. Determine the electron-domain geometry using VSEPR model
3. Specify hybrid orbitals for the bonding electron pairs based on their geometric arrangement



Hybrid Orbital Geometries

Atomic Orbital Set	Hybrid Orbital Set	Geometry	Examples
s,p	Two *sp*	 Linear	BeF_2, $HgCl_2$
s,p,p	Three sp^2	 Trigonal planar	BF_3, SO_3
s,p,p,p	Four sp^3	 Tetrahedral	CH_4, NH_3, H_2O, NH_4^+

Hybrid Orbital Geometries

Atomic Orbital Set	Hybrid Orbital Set	Geometry	Examples
s,p,p,p,d	Five sp^3d	 Trigonal bipyramidal	PF_5, SF_4, BrF_3
s,p,p,p,d,d	Six sp^3d^2	 Octahedral	SF_6, ClF_5, XeF_4, PF_6^-

The center atom of CO_2 adopts following orbital hybridization:

A) sp

B) sp^2

C) sp^3

D) sp^3d

The center atom of PH_3 adopts following orbital hybridization:

A) sp

B) sp^2

C) sp^3

D) sp^3d

The center atom of PCl_5 adopts following orbital hybridization:

A) sp^3

B) sp^3d

C) sp^3d^2

D) sp^3d^3

The center atom of $POCl_3$ adopts following orbital hybridization:

A) sp^3

B) sp^3d

C) sp^3d^2

D) sp^3d^3

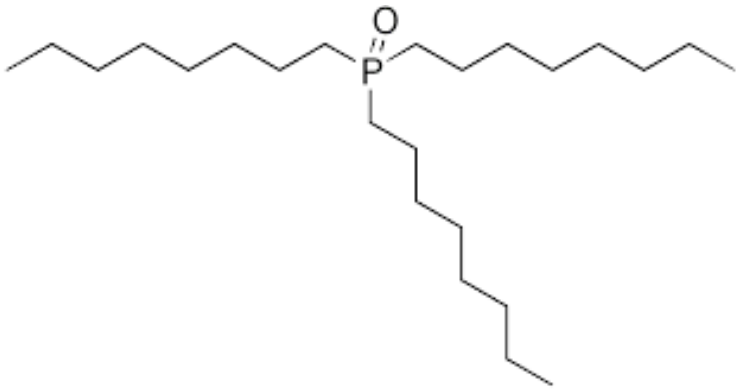
The hybridization of phosphours in $POCl_3$ is the same as in:

A) P in PCl_3

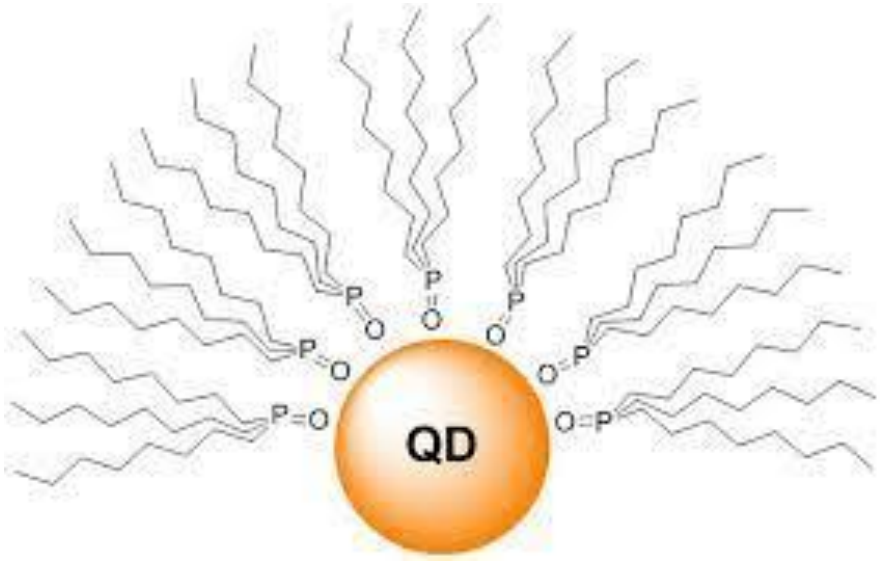
B) S in SF_4

C) B in BCl_3

D) C in CS_2

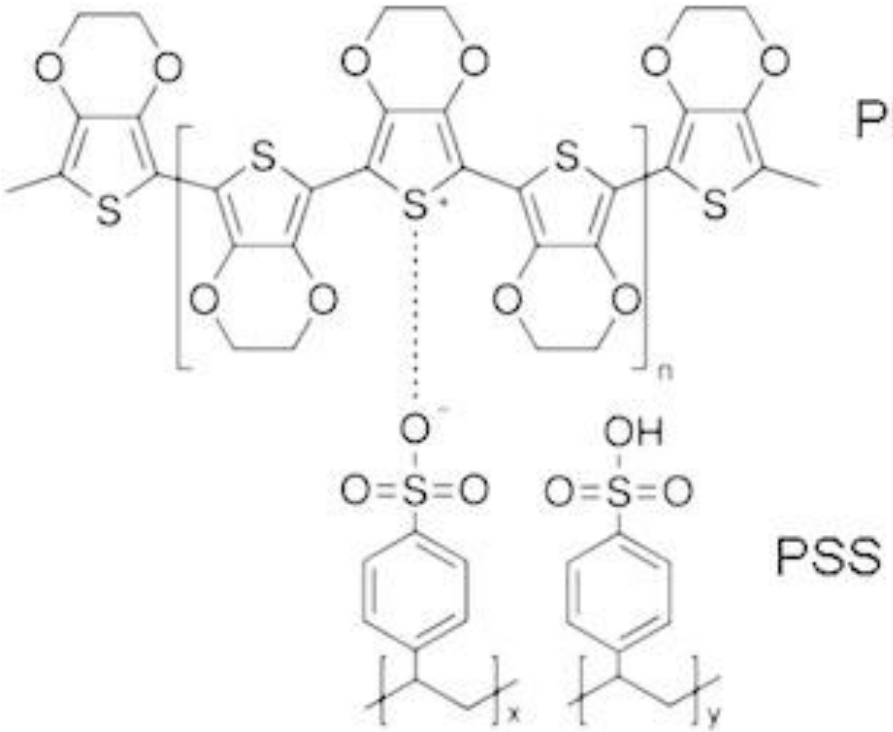


TOPO = TriOctyl Phosphine Oxide



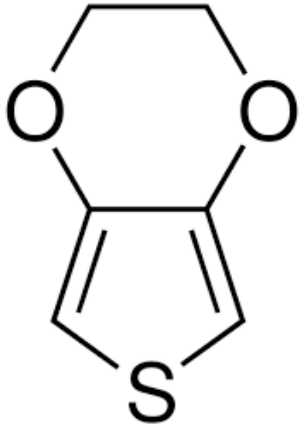
Quantum dots with TOPO surfactant

What is hybrid orbital (orbital hybridization) of the S atom within EDOT?



Conducting polymer

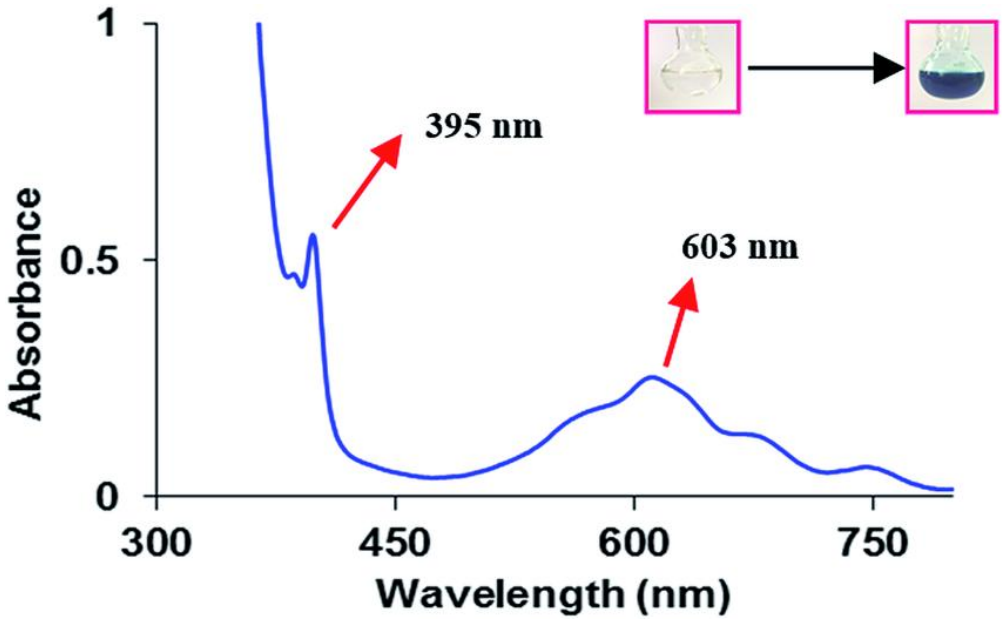
Oxidation



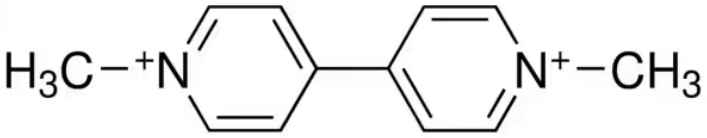
3,4-Ethylenedioxythiophene

(EDOT)

What is hybrid orbital (orbital hybridization) of the N atom within Paraquat?



Electron transfer mediator

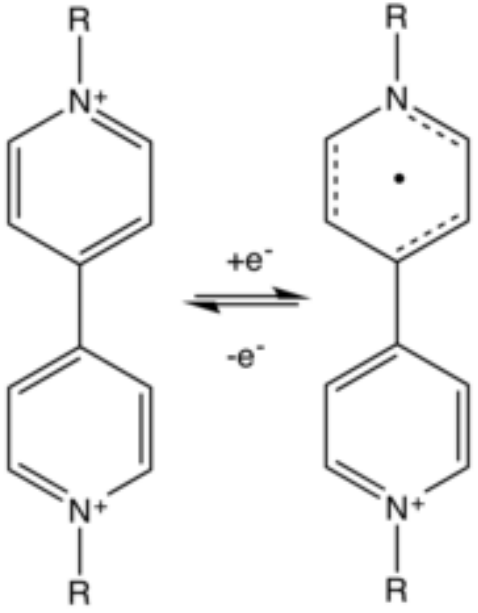


$2Cl^- \cdot xH_2O$

Methyl viologene

(Paraquat)

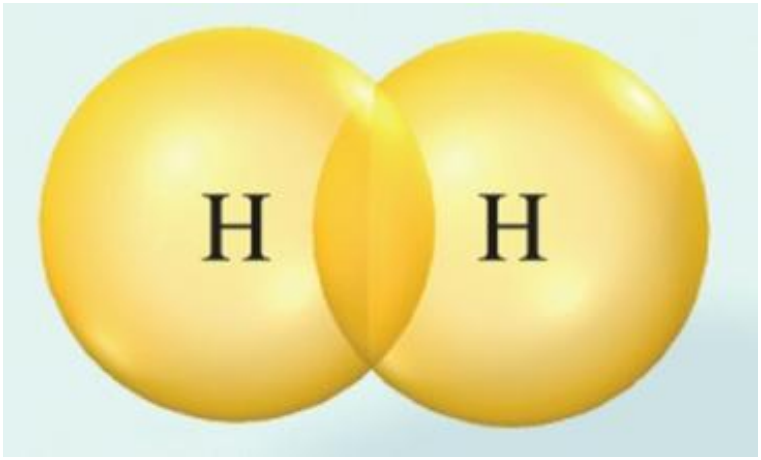


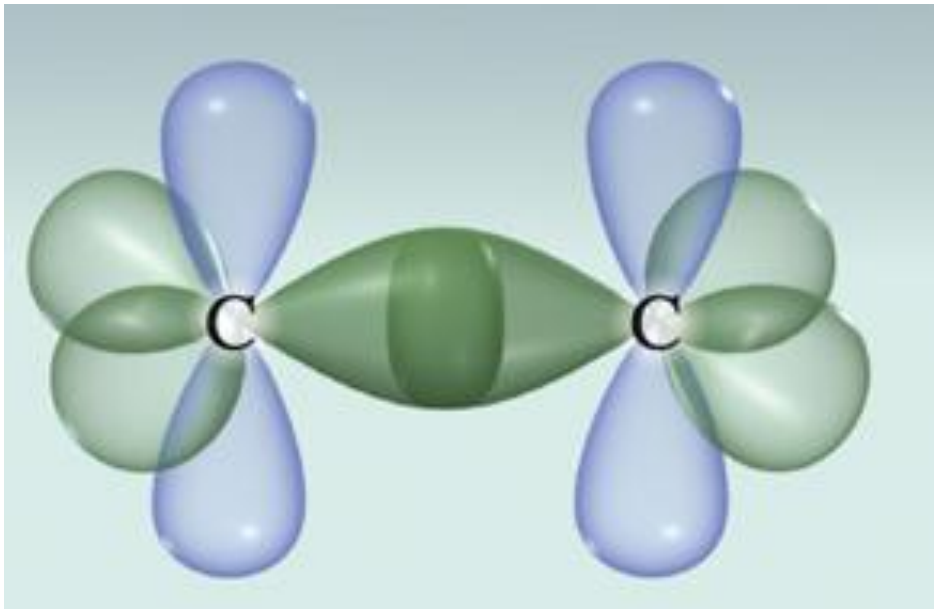


6. Multiple bonds

Sigma (σ) bond

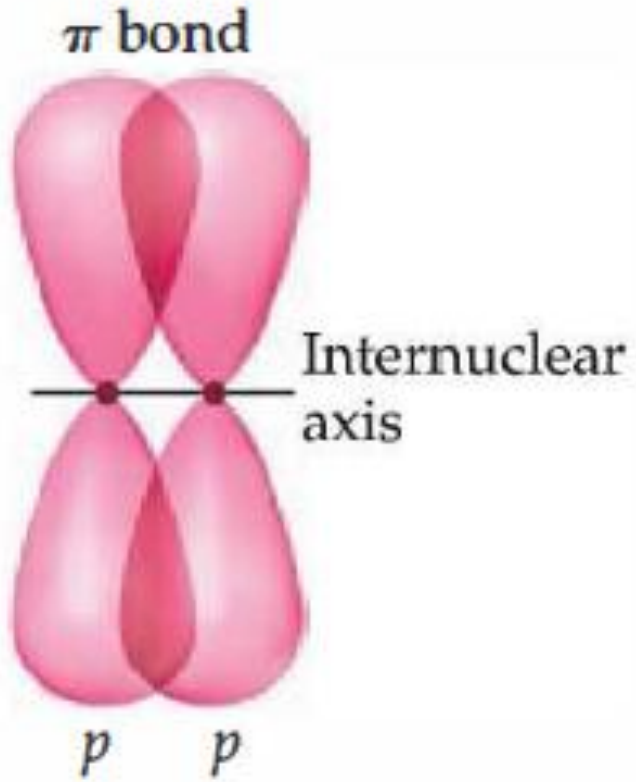
*A **sigma (σ) bond*** is a bond in which the shared electron density is concentrated directly along the internuclear axis between the two nuclei involved in bonding.





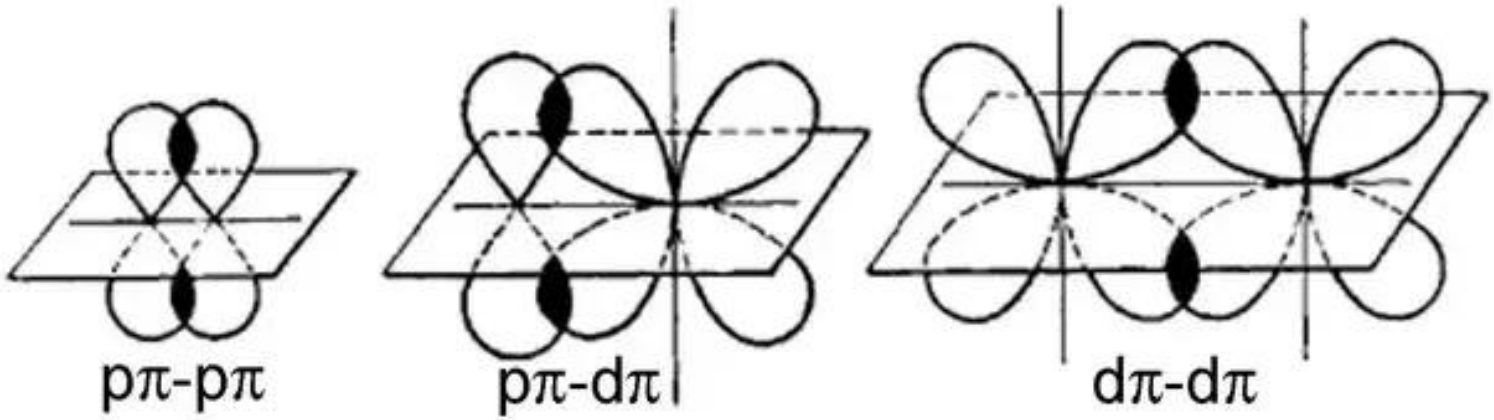
Pi (π) *bond*

*A **pi** (π) **bond*** forms from the sideways overlap of ***p*** orbitals resulting in regions of electron density that are concentrated above and below the plane of the molecule.



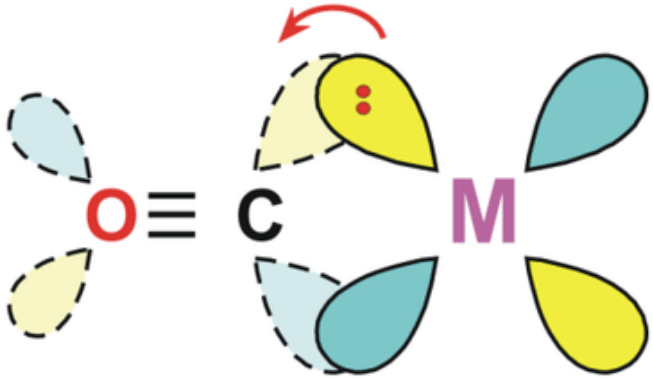
Pi (π) *bond*

*A **pi** (π) **bond*** forms from the sideways overlap of ***p*** *and/or* ***d*** orbitals resulting in regions of electron density that are concentrated above and below the plane of the molecule.





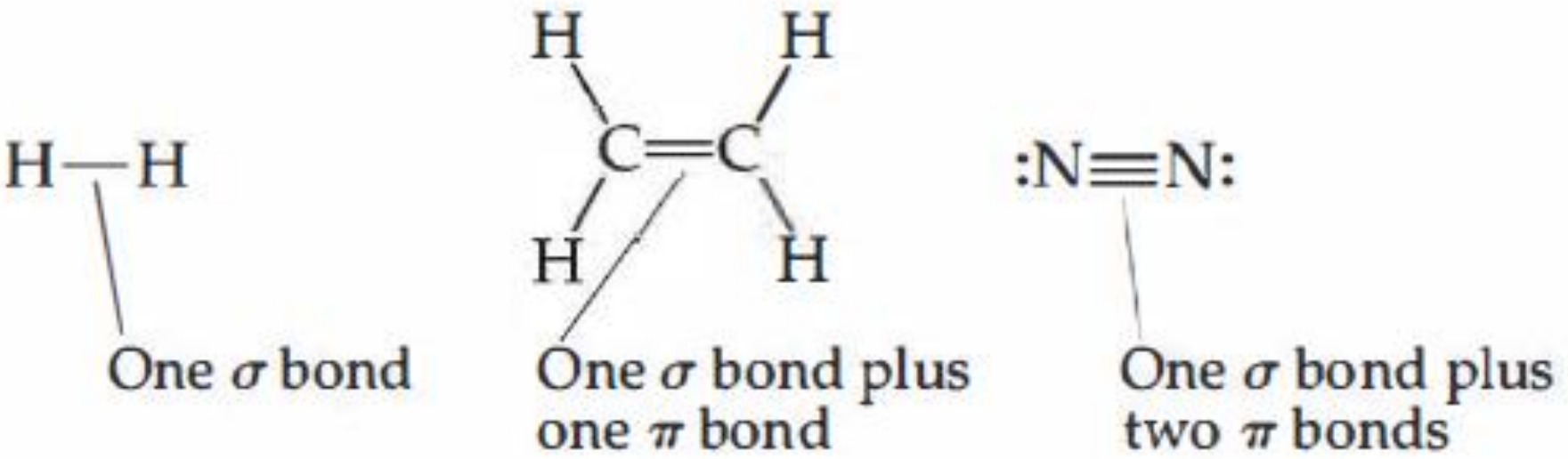
CO-M sigma bond



M to CO pi backbonding

Multiple bonds

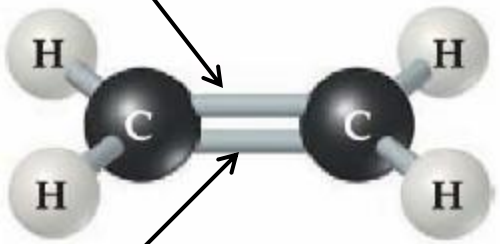
- Single bonds are δ bond
- A double bond = 1 δ bond + 1 π bond
- A triple bond = 1 δ bond + 2 π bonds



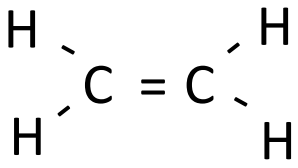
Double bonds in C_2H_4

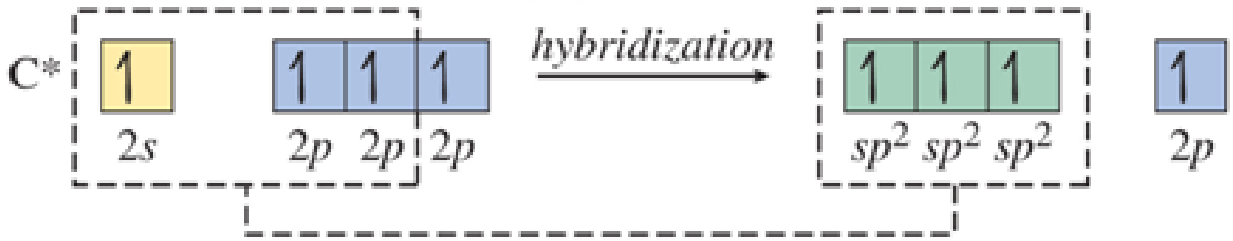
δ bond

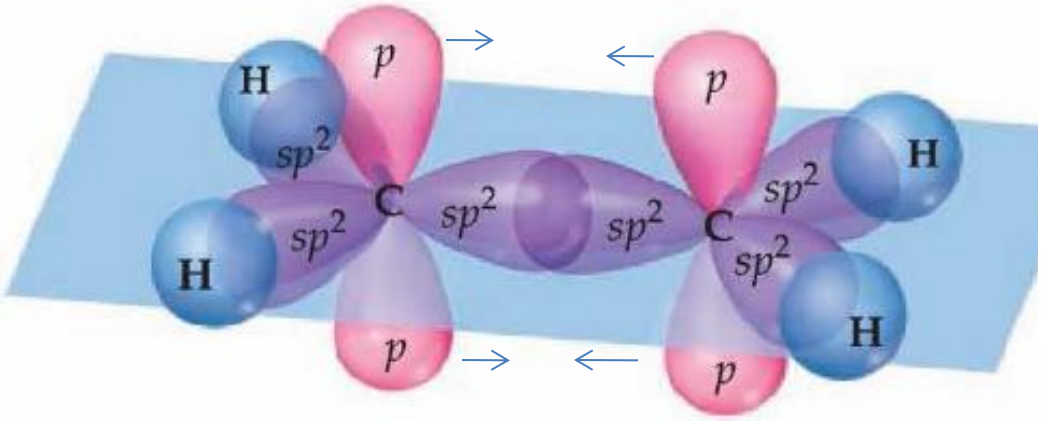
π bond



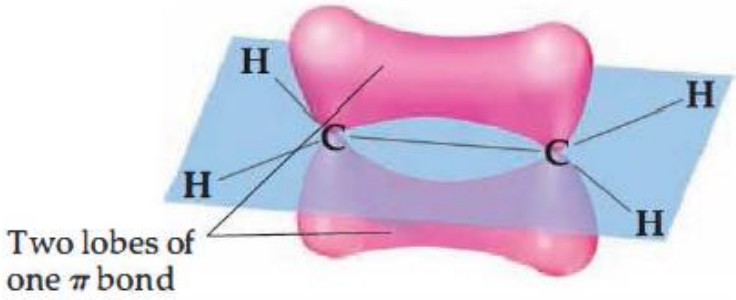
number of = 3
electron domains







δ bond



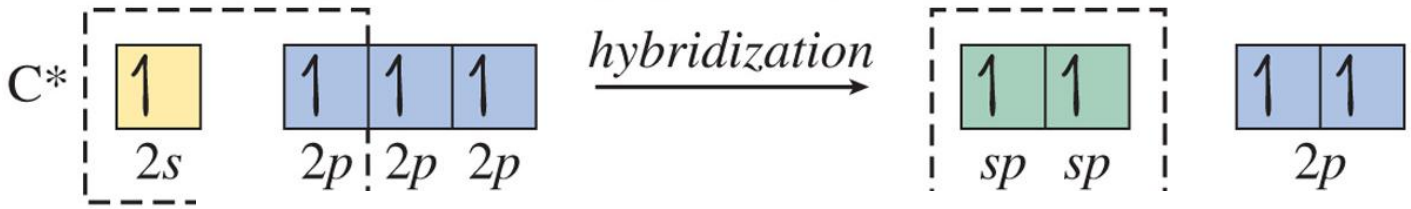
π bond

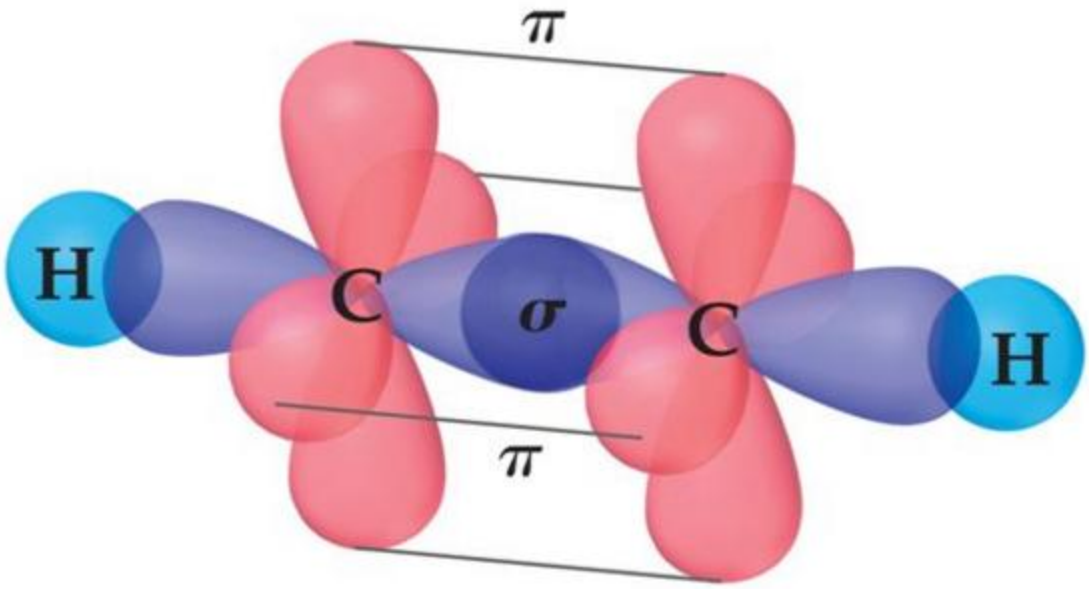
Triple bonds in C_2H_2

Acetylene, C_2H_2 $\quad H-C\equiv C-H$

number of electron domains = 2

hybridization = *sp*



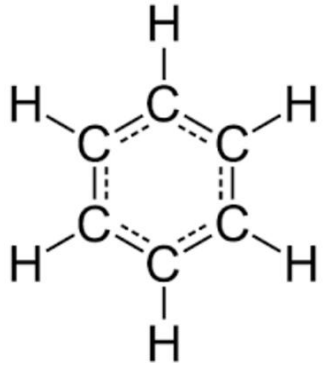


Resonance structures

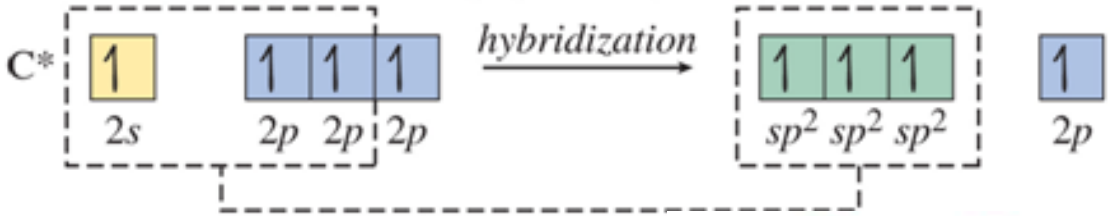
Benzene C_6H_6 resonance structures



number of electron domains = 3

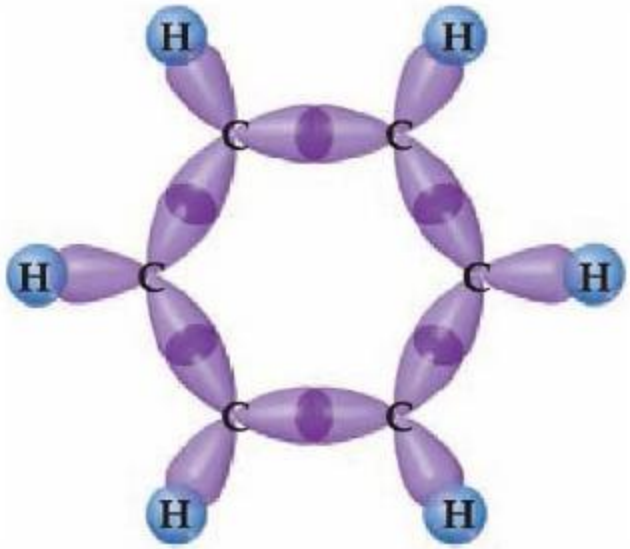


Hybrid orbitals: sp^2





(b) 2p atomic orbitals



(a) σ bonds

Benzene Resonance Structures



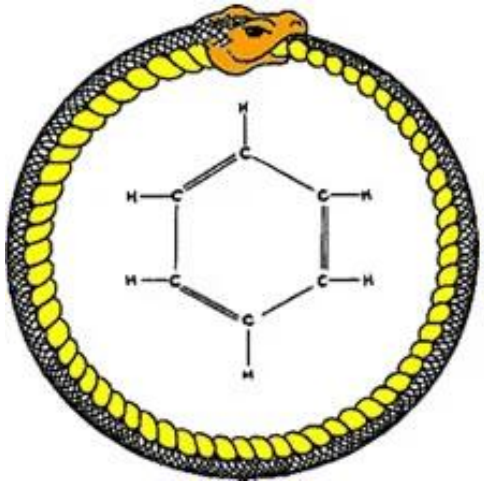
(a) Localized π bonds



(b) Localized π bonds



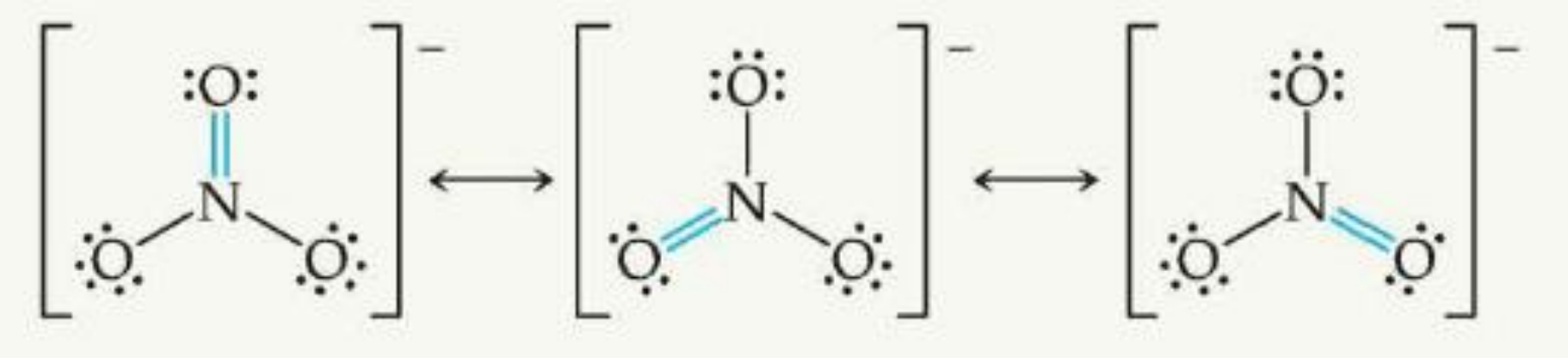
(c) Delocalized π bonds

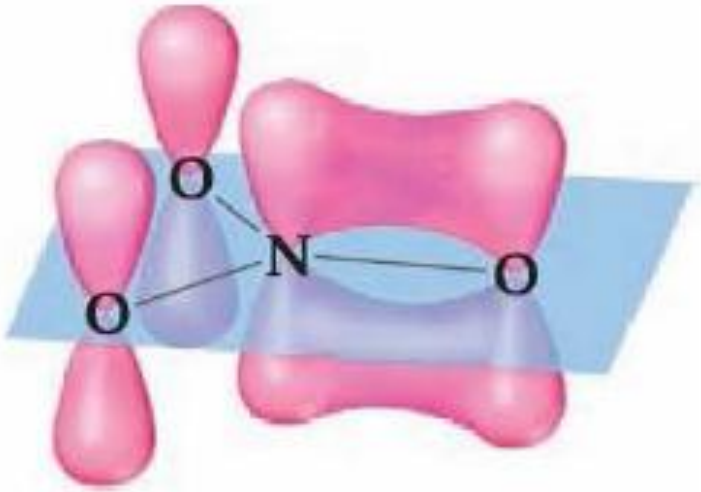




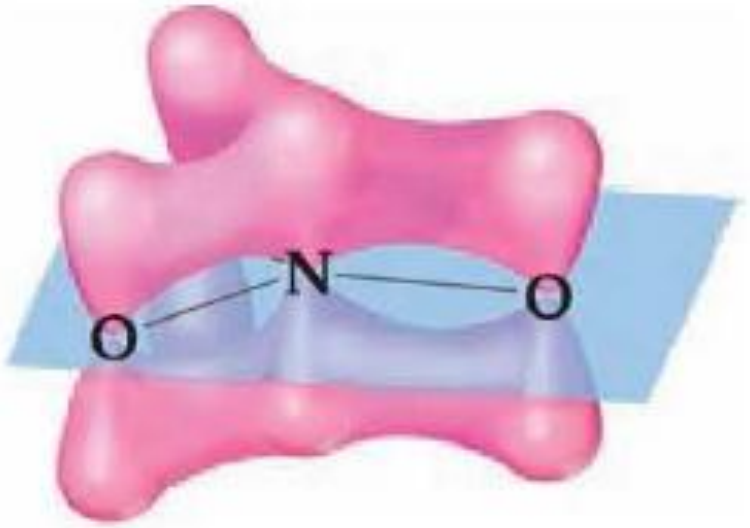
August Kekule (1829-1896)

NO_3^- Resonance Structure





(a) N—O π bond in one of the resonance structures of NO_3^-.



(b) Delocalization of the π bonds in the NO_3^- ion.

BESSY II Synchrotron @ Berlin



The nitrogen atoms in N_2 participate in multiple bonding, whereas those in hydrazine, N_2H_4, do not.

A) Draw Lewis structures for both molecules.

B) What is the hybridization of the nitrogen atoms in each molecule?

C) Which molecule has the stronger N ¬ N bond?

A) Write a single Lewis structure for SO_3, and determine the hybridization at the S atom.

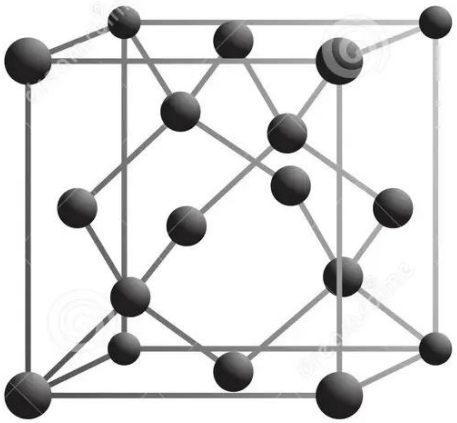
B) Are there other equivalent Lewis structures for the molecule?

C) Would you expect SO_3 to exhibit delocalized p bonding?

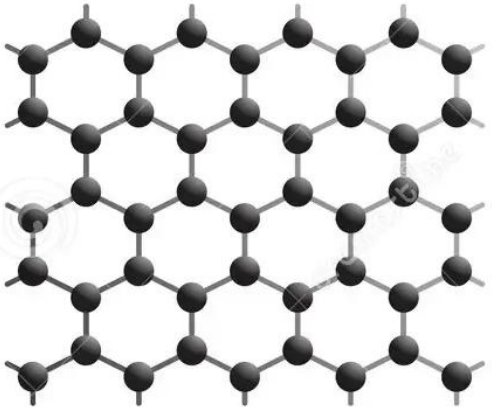
A) Determine the hybridization at the C atom within diamond and graphene.

B) In which compound (above two carbon allotropes) would you expect to have delocalized p bonding?

C) Which compound would offer higher electrical conductivity?



Diamond



Graphene

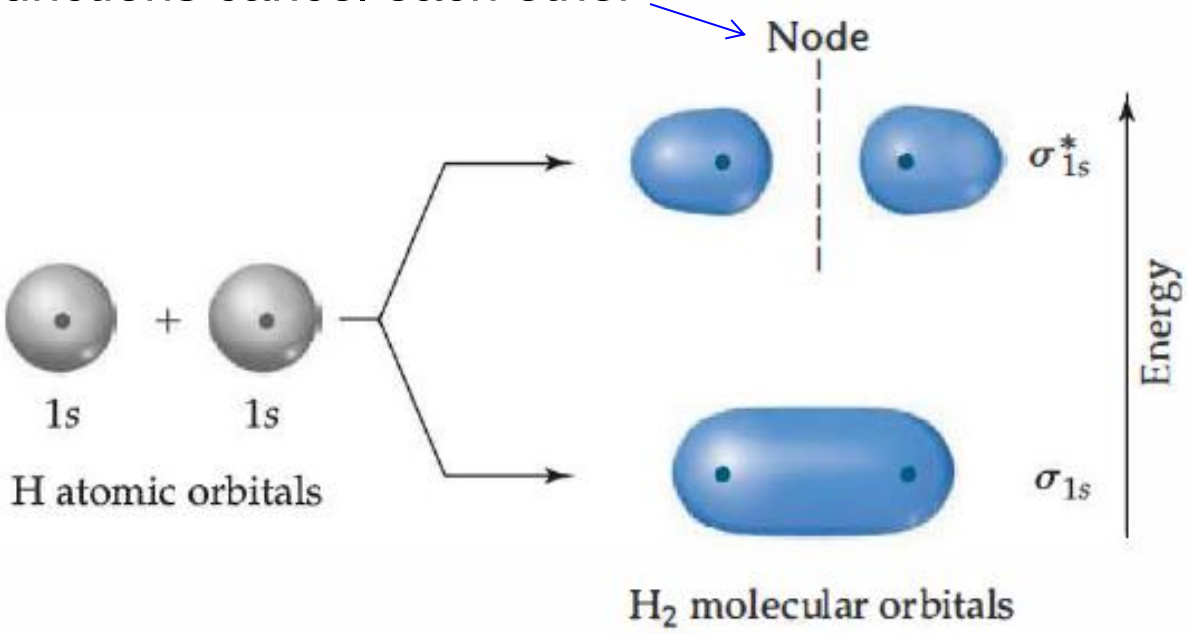
7. Molecular Orbitals

- Valence bond theory and hybrid orbitals are inadequate to describe the excited state of molecules
- **Molecular orbital theory**: describes the electrons in molecules by using specific wave functions called **molecular orbitals** or **MO**
- MOs are associated with the entire molecule, not with a single atom

The Hydrogen Molecule

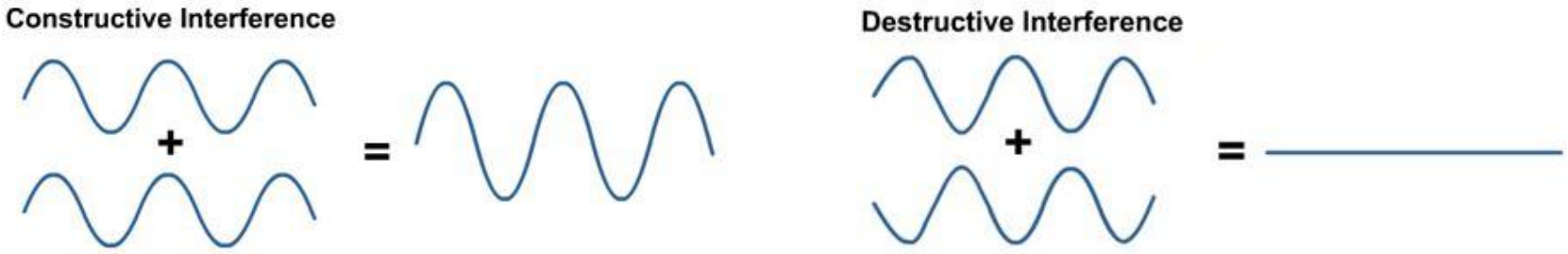
When 2 atomic orbitals overlap, two molecular orbitals form

Wave functions cancel each other

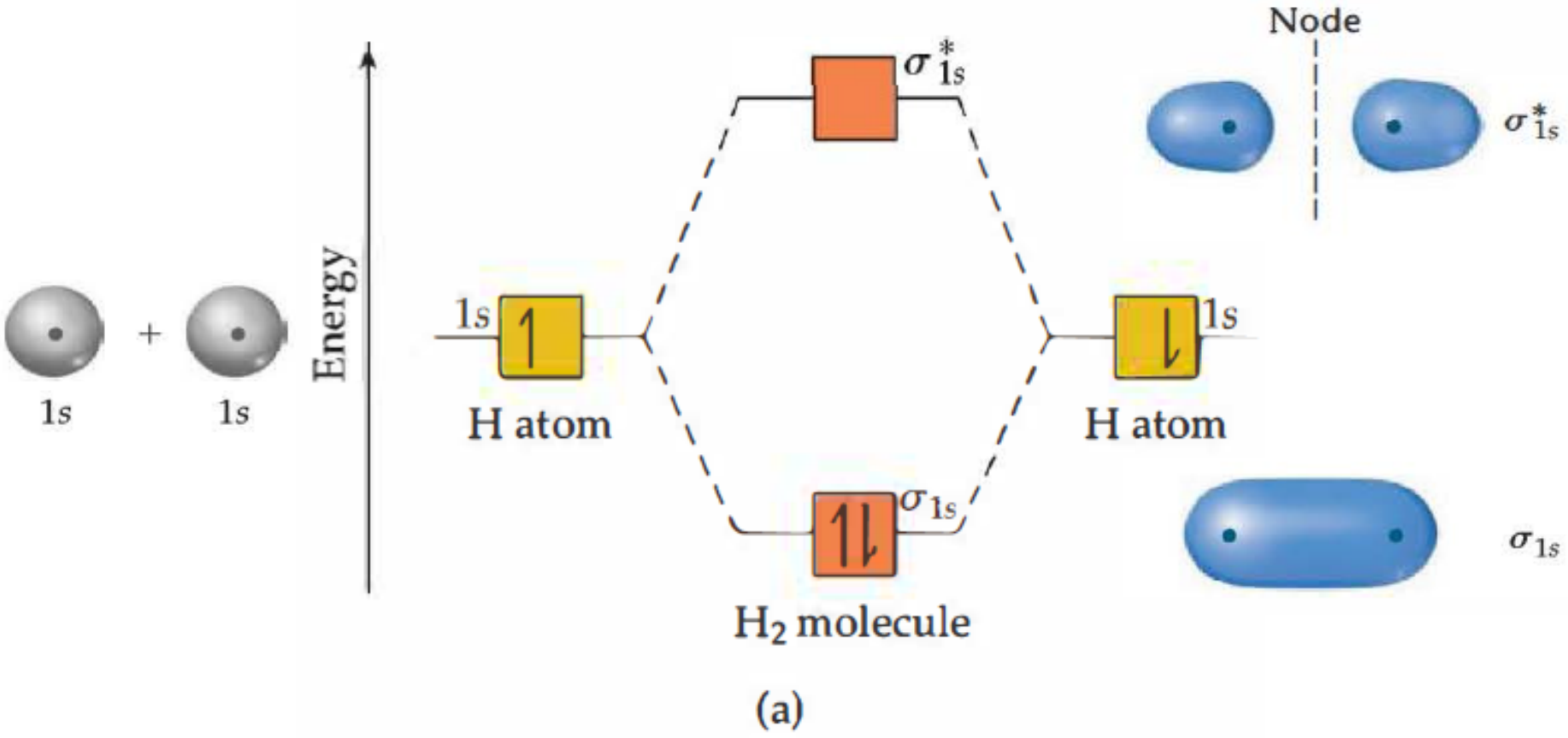


Antibonding
molecular orbital

Bonding
molecular orbital



H_2 energy level diagram

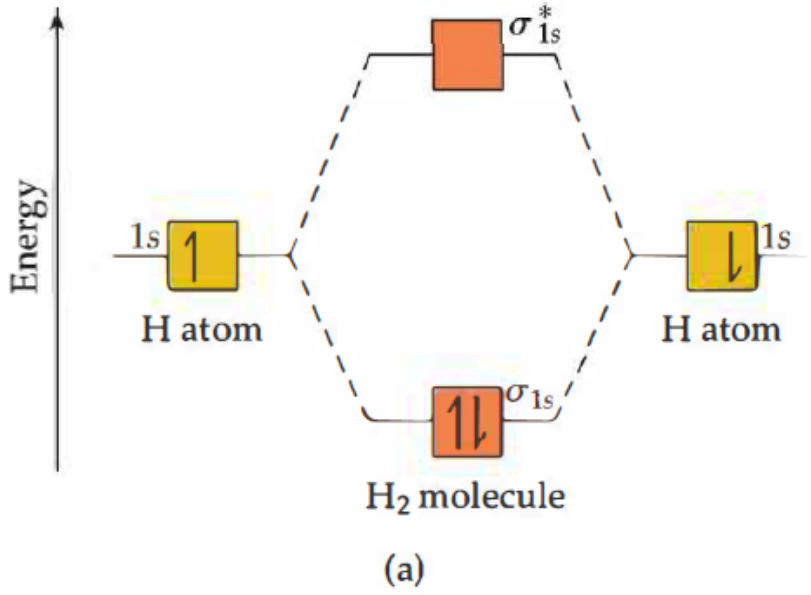


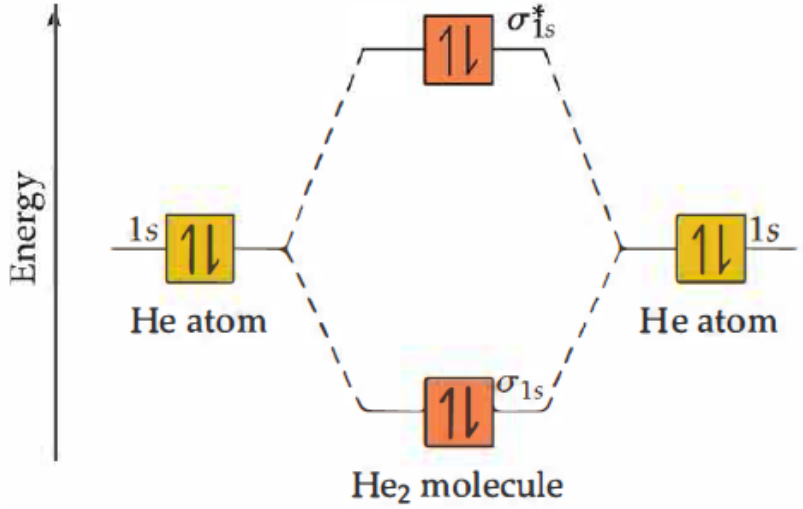
The two electrons occupy the lower energy bonding δ_{1s} MO with lower energy than the isolated 1s atomic orbitals

Bond order

- Bond order: stability of a covalent bond

$$\text{bond order} = \frac{\text{number of electrons in bonding molecular orbitals} - \text{number of electrons in antibonding molecular orbitals}}{2}$$

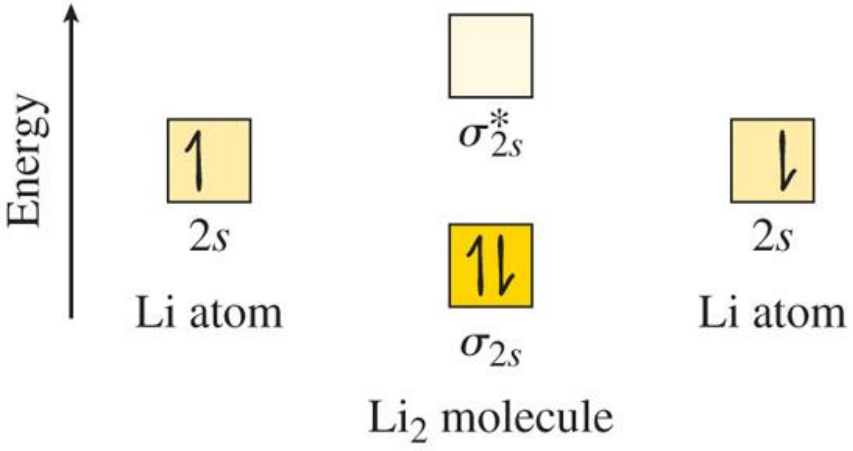




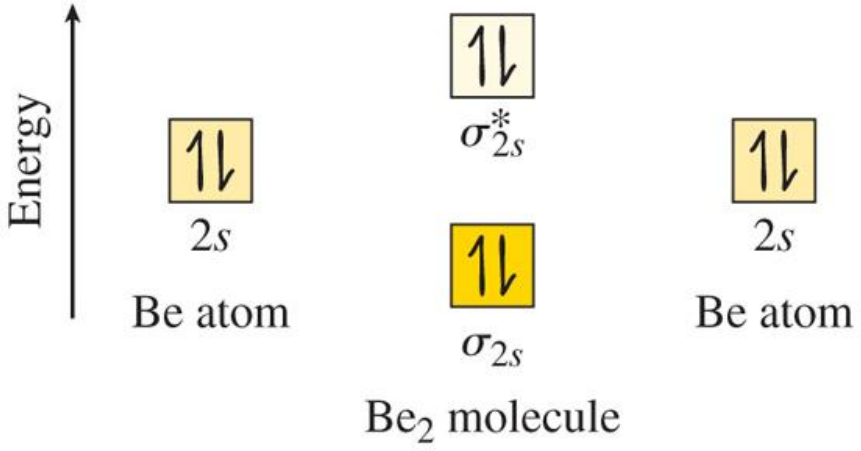
bond order = 1/2 (2 – 0) = 1
single bond

bond order = 1/2 (2 – 2) = 0
no bond

Bond order

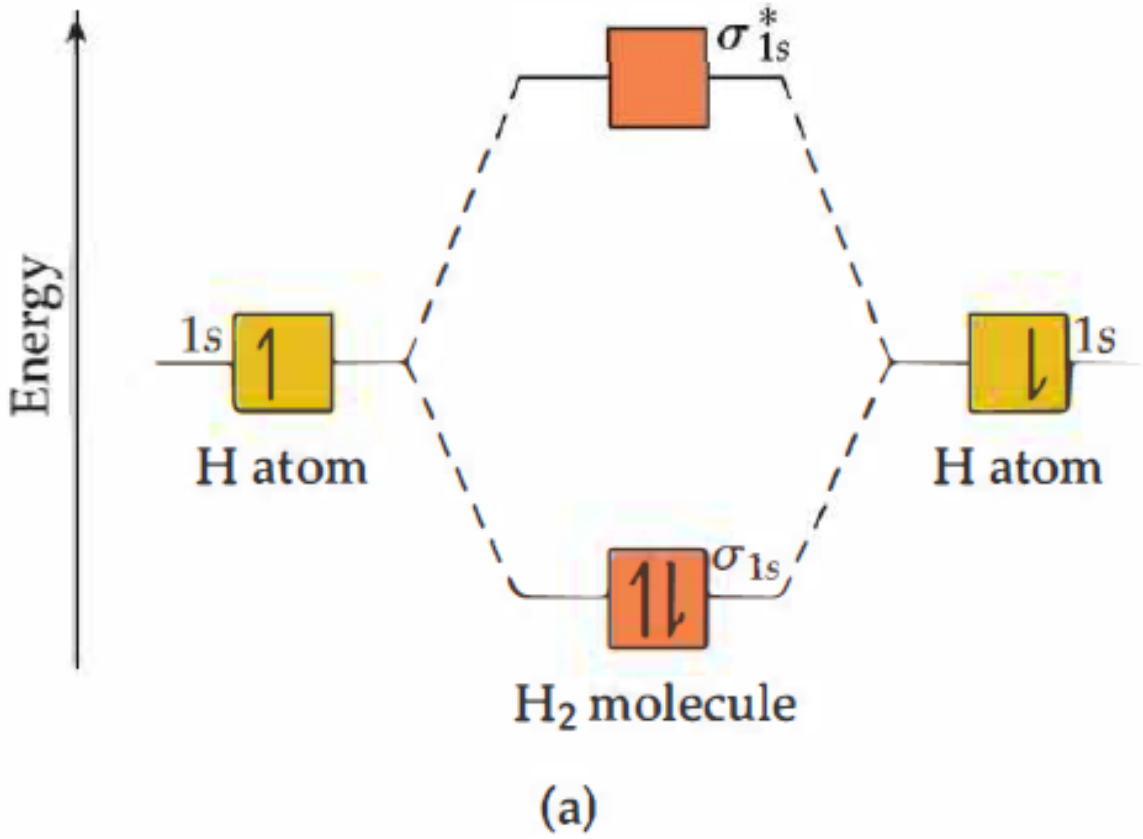


bond order = 1/2 (2 – 0) = 1
single bond



bond order = 1/2 (2 – 2) = 0
no bond

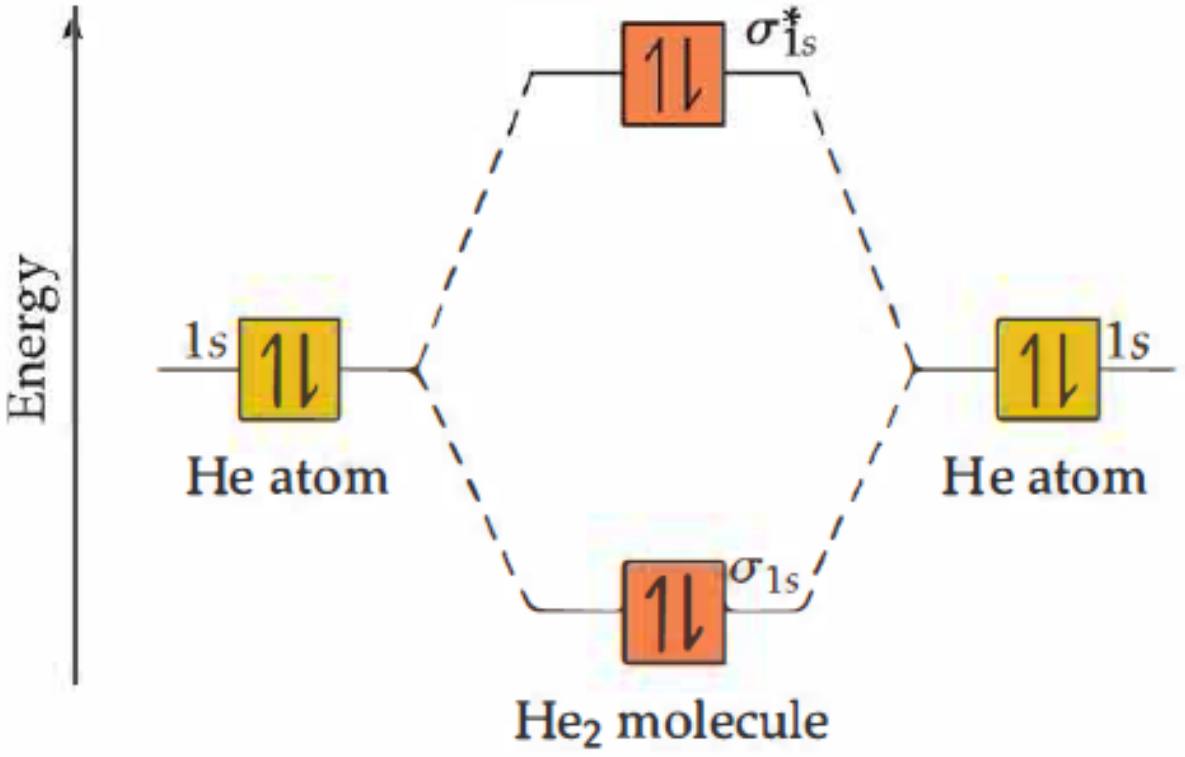
H_2 energy level diagram



The two electrons occupy the lower energy bonding δ_{1s} MO with lower energy than the isolated 1s atomic orbitals

H_2 is more stable than H atom

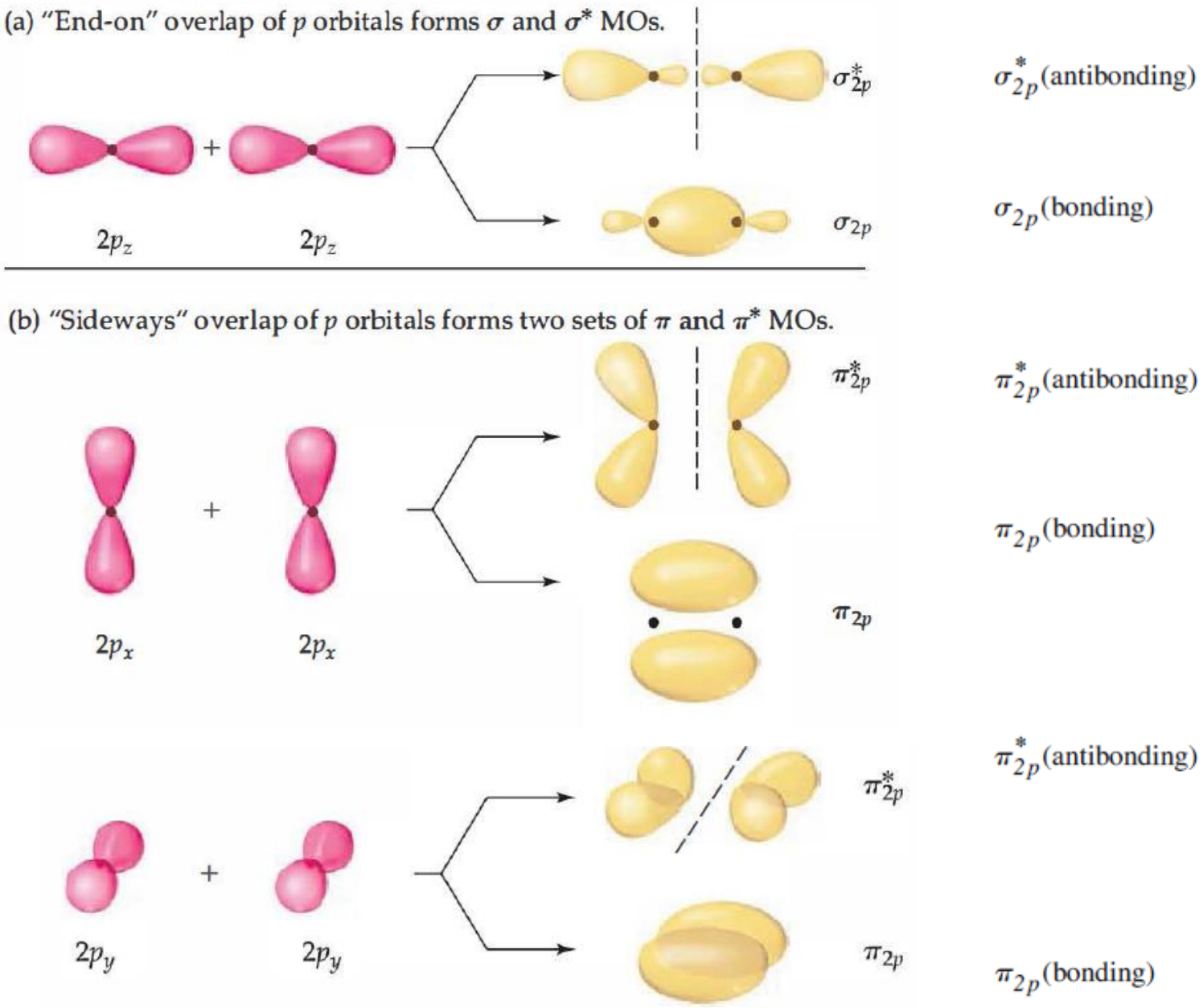
He energy level diagram



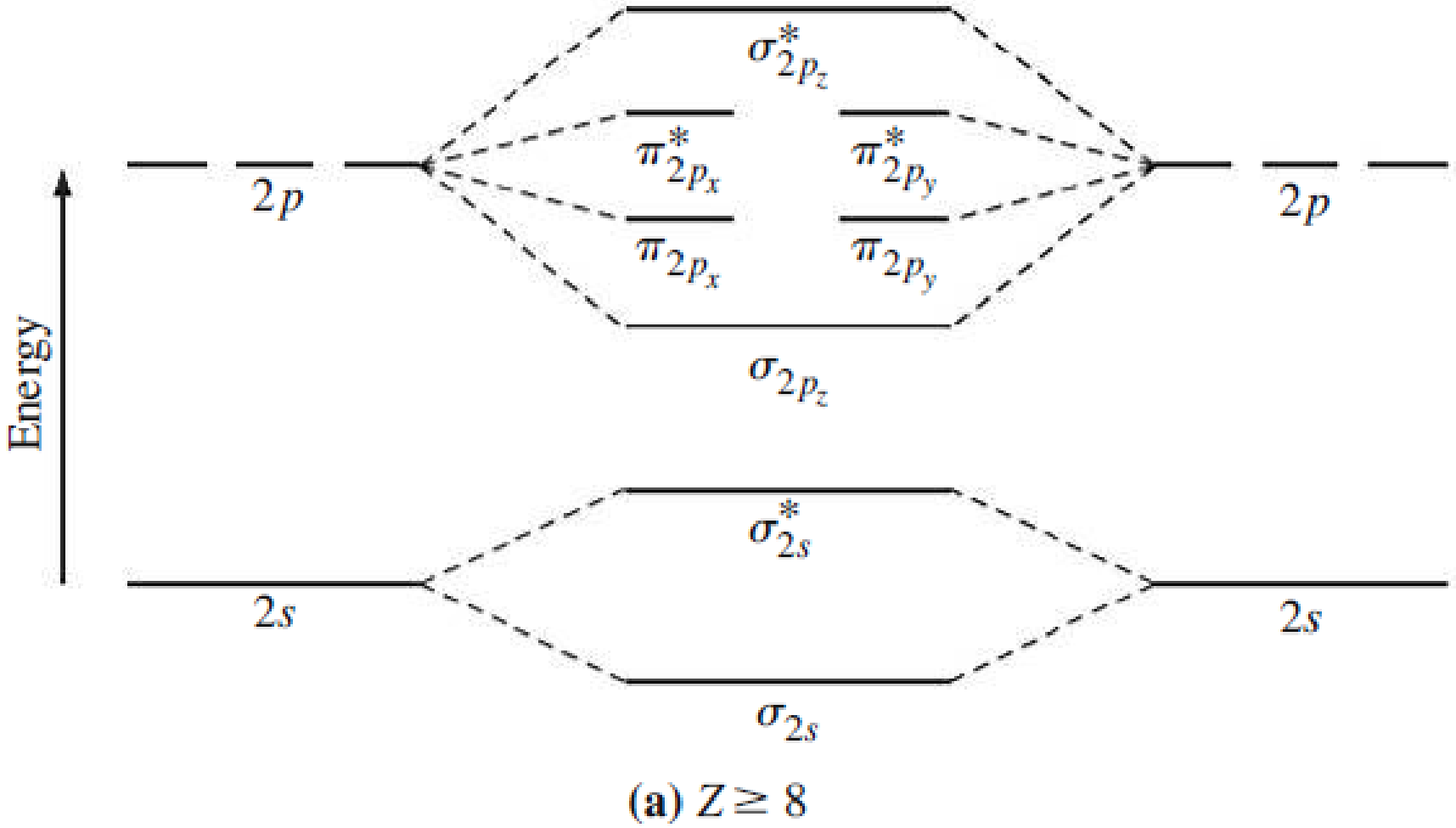
The combination of energy in δ^*_{1s} and δ_{1s} MOs are equivalent to energy in 1s atomic orbitals

He_2 is not more stable than He atom

Molecular Orbitals from 2p Atomic Orbitals

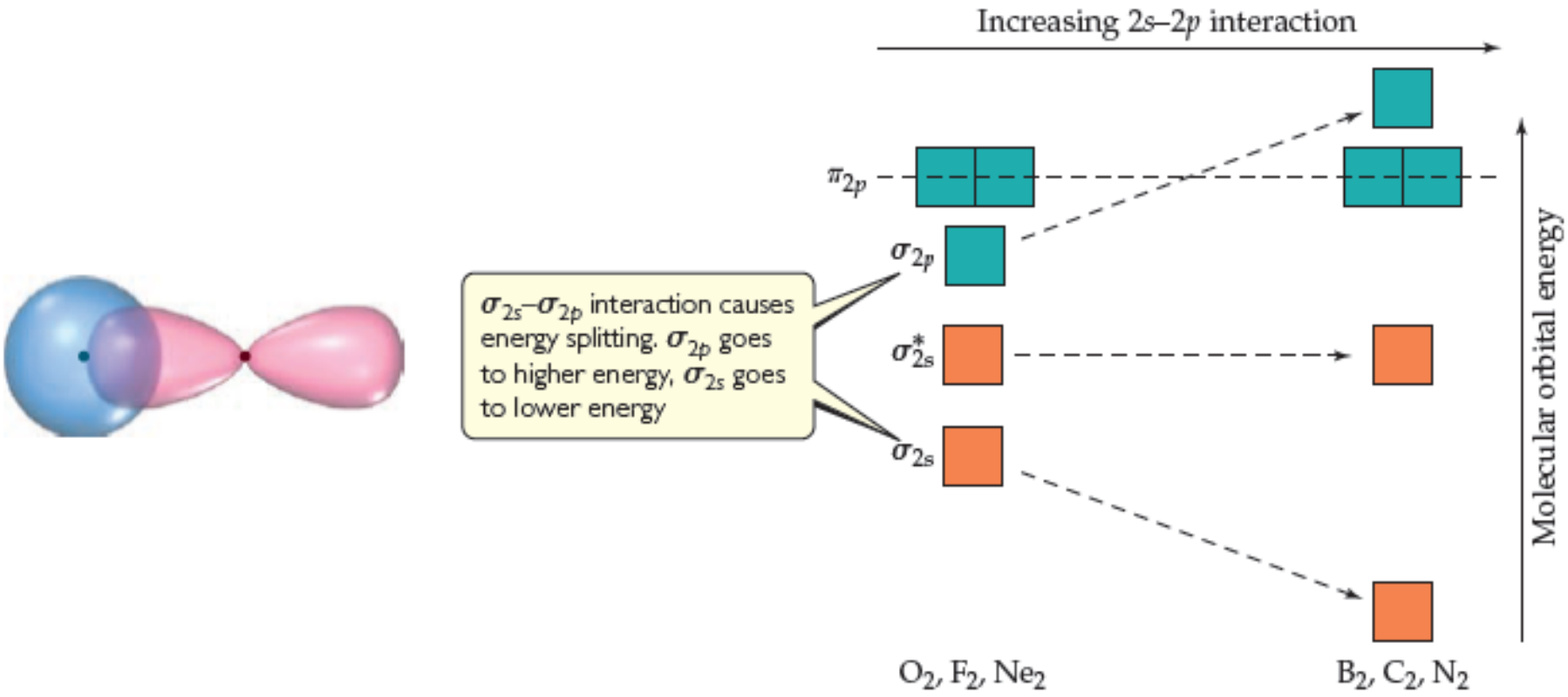


Energy Levels of 2s and 2p MOs



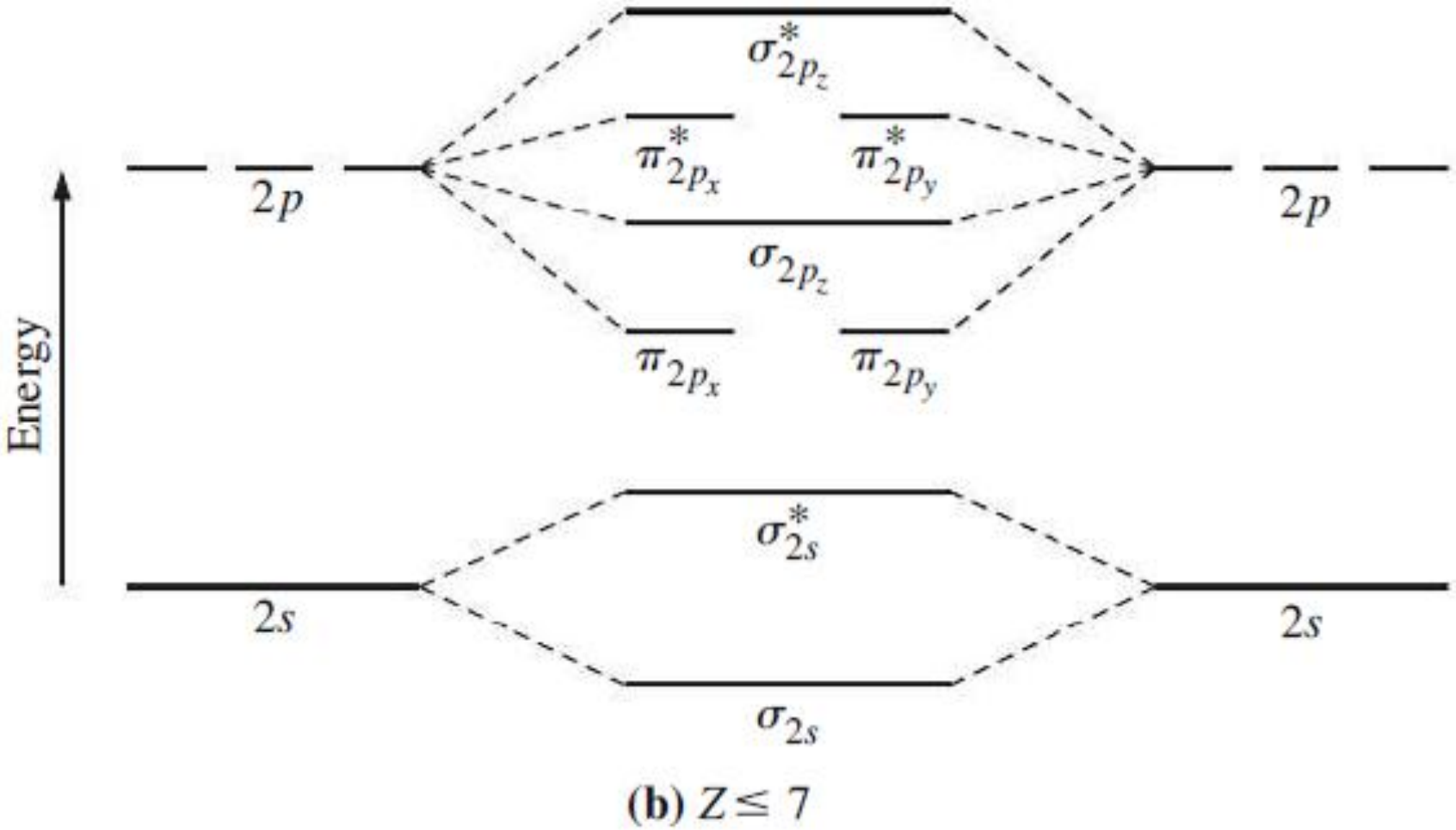
(a) $Z \geq 8$

s–p interaction causes energy spliting



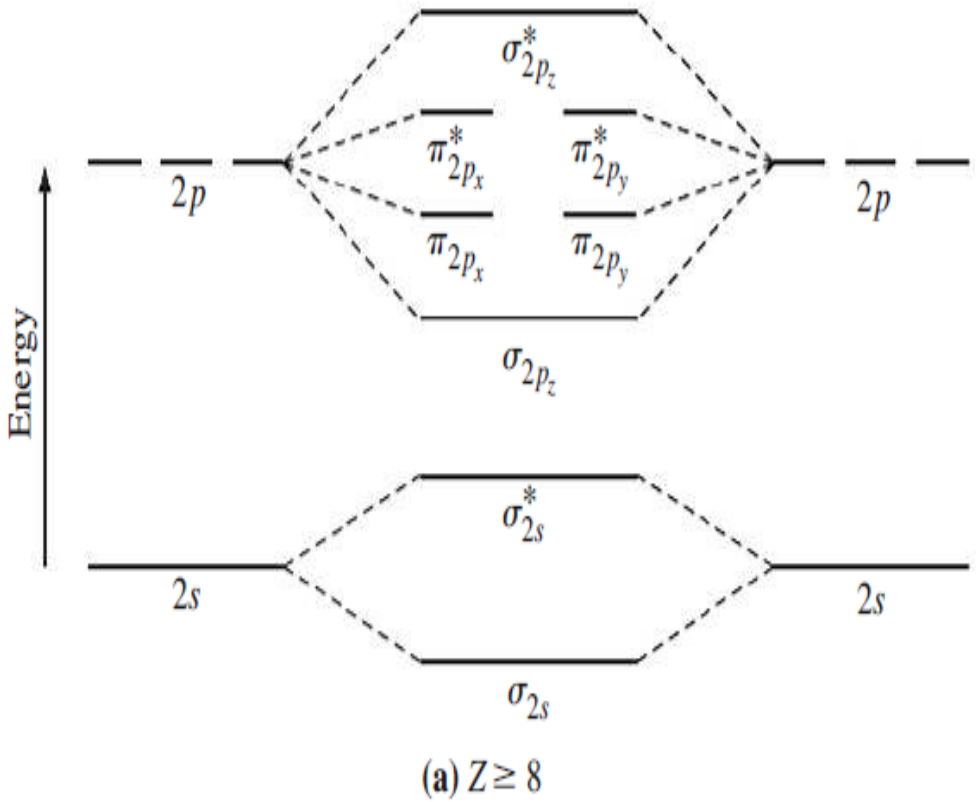
▲ **Figure 9.42** The effect of interactions between 2*s* and 2*p* atomic orbitals.

Energy Levels of 2s and 2p MOs

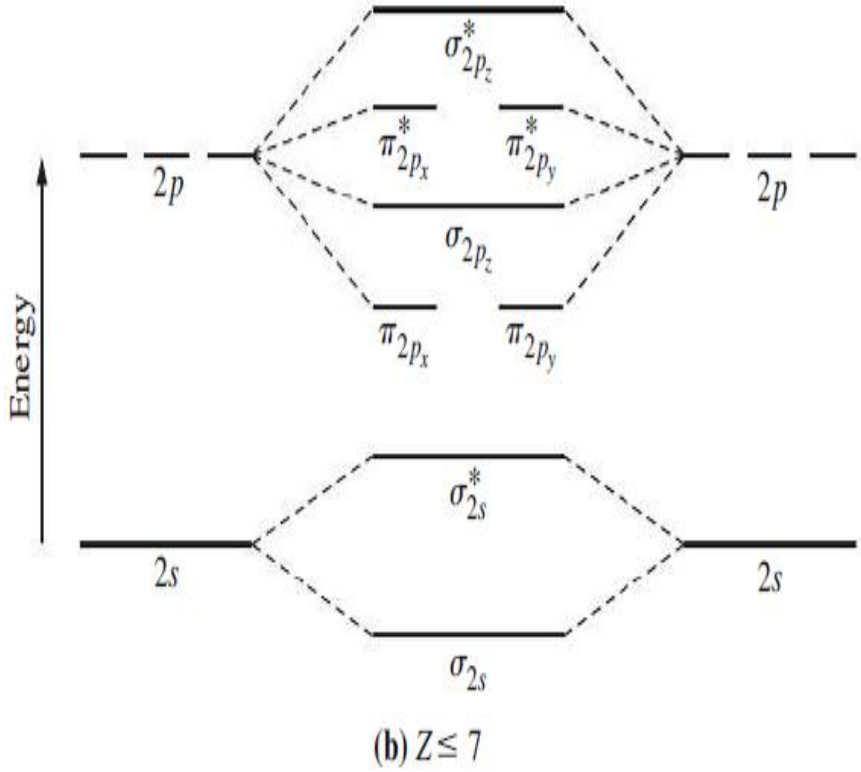


(b) $Z \leq 7$

Energy Levels of 2s and 2p MOs



(a) $Z \geq 8$



(b) $Z \leq 7$

Magnetism

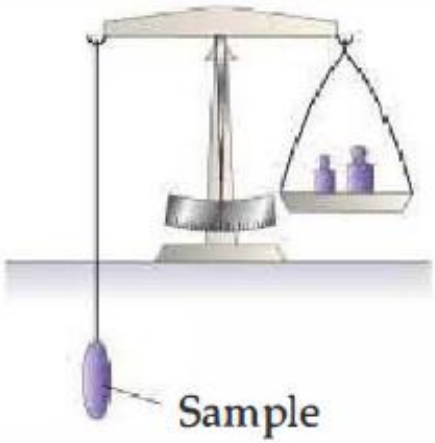
– ***Diamagnetic substance***

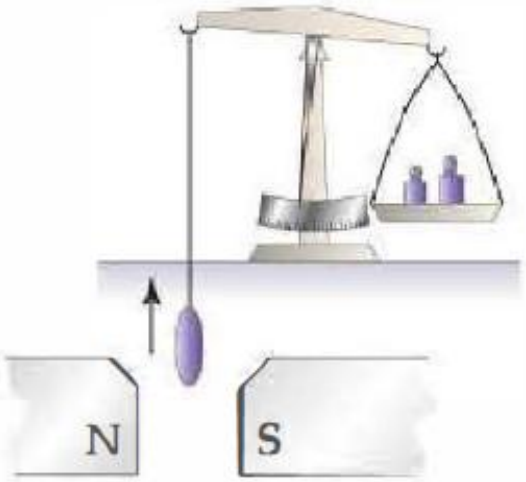
- A substance whose electrons are all paired.
- ↑↓
- Weakly repelled by magnetic fields.

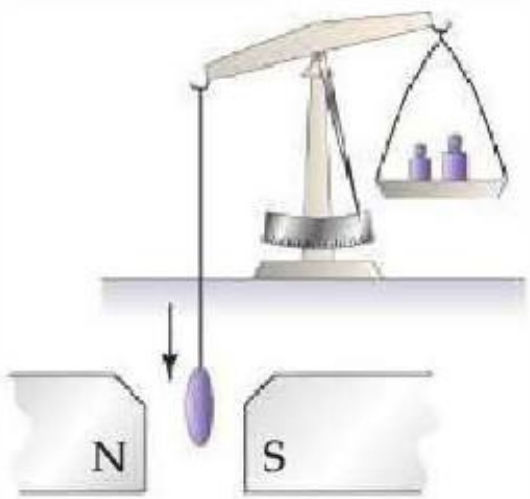
– ***Paramagnetic substance***

- A substance with one or more unpaired electrons.
- ↑
- Attracted by magnetic fields.

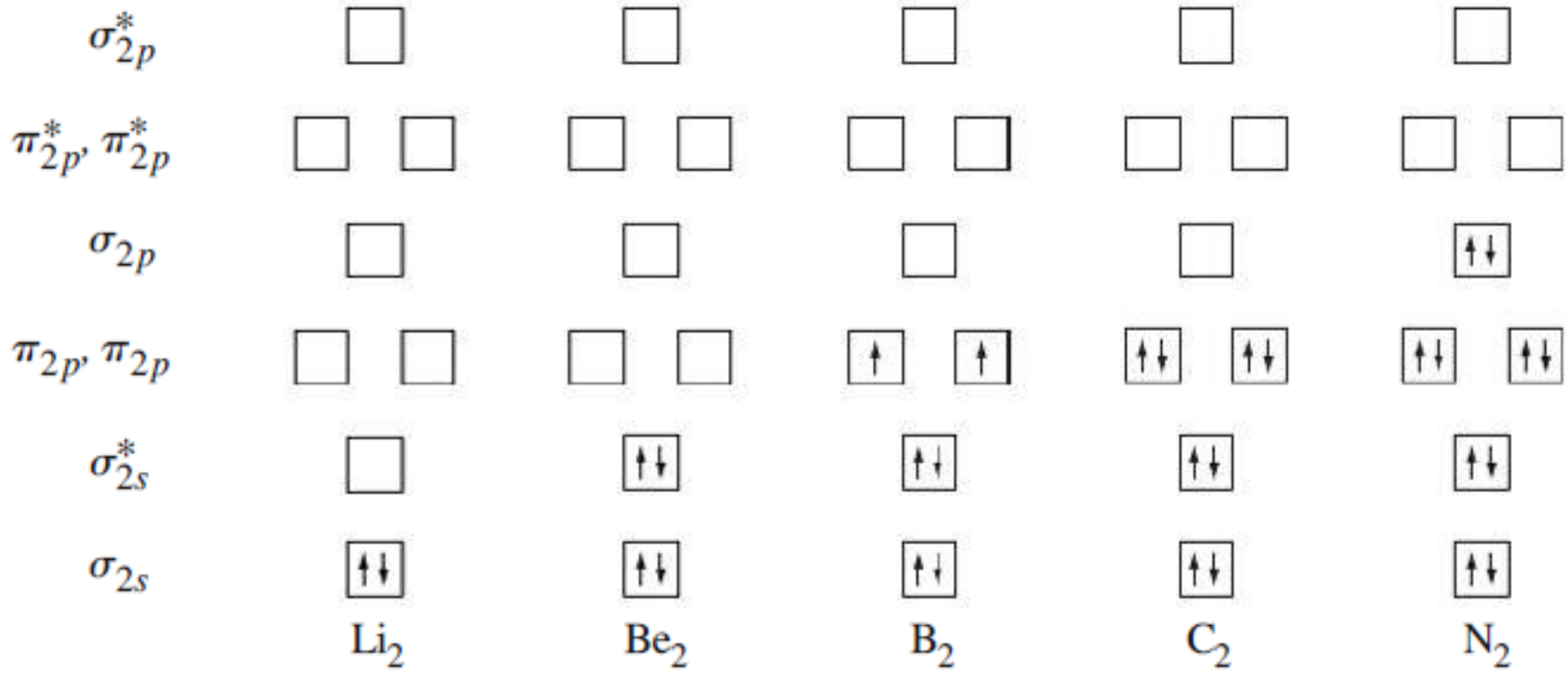
Determine the Magnetic Properties





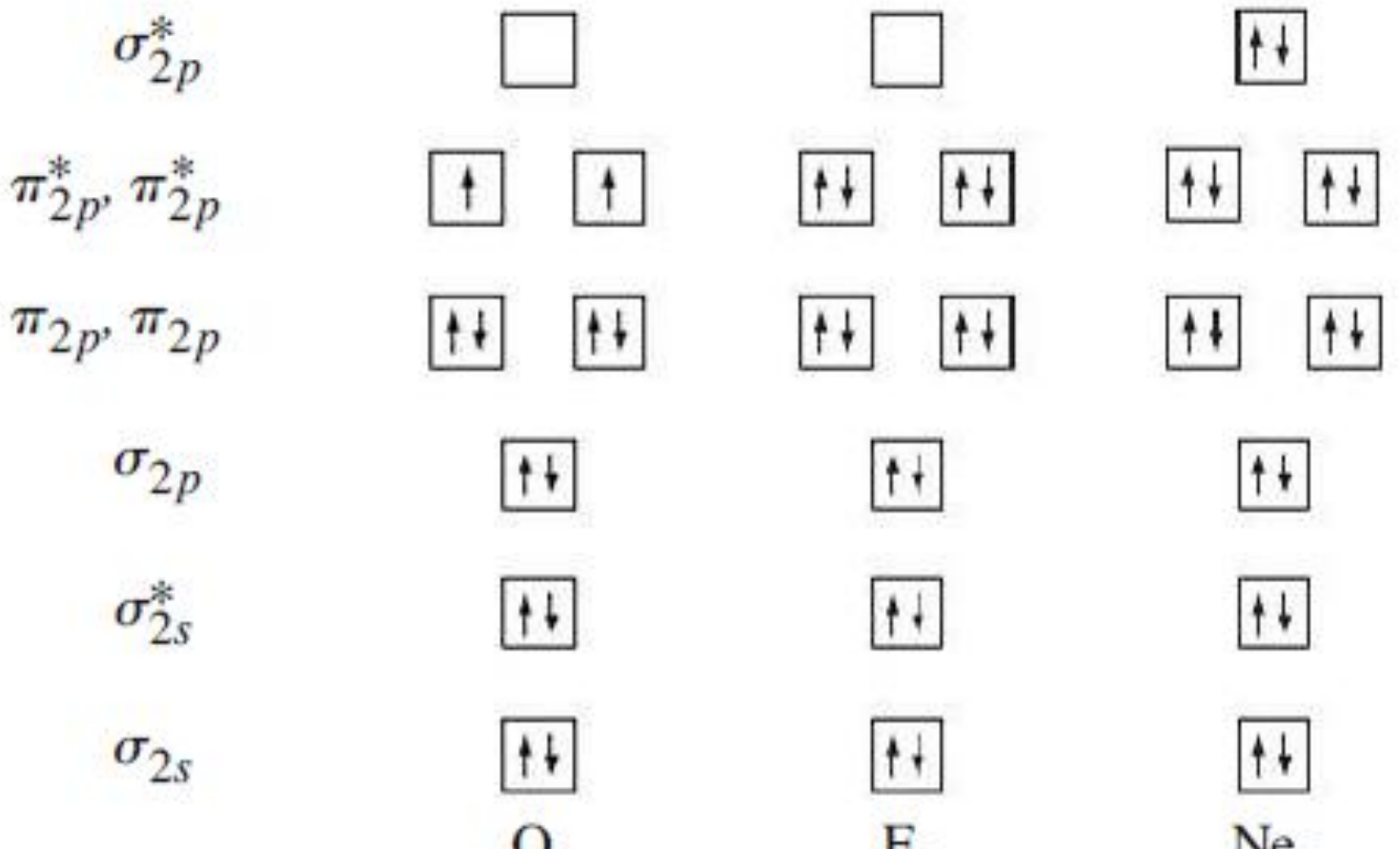


Second-row Diatomic Molecules



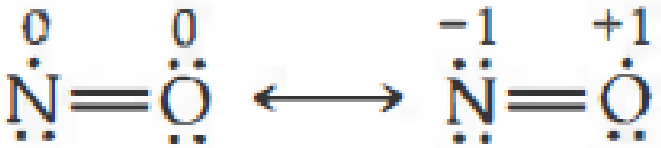
	Li_2	Be_2	B_2	C_2	N_2
Bond order	1	0	1	2	3
Magnetism	Diamagnetic	–	Paramagnetic	Diamagnetic	Diamagnetic

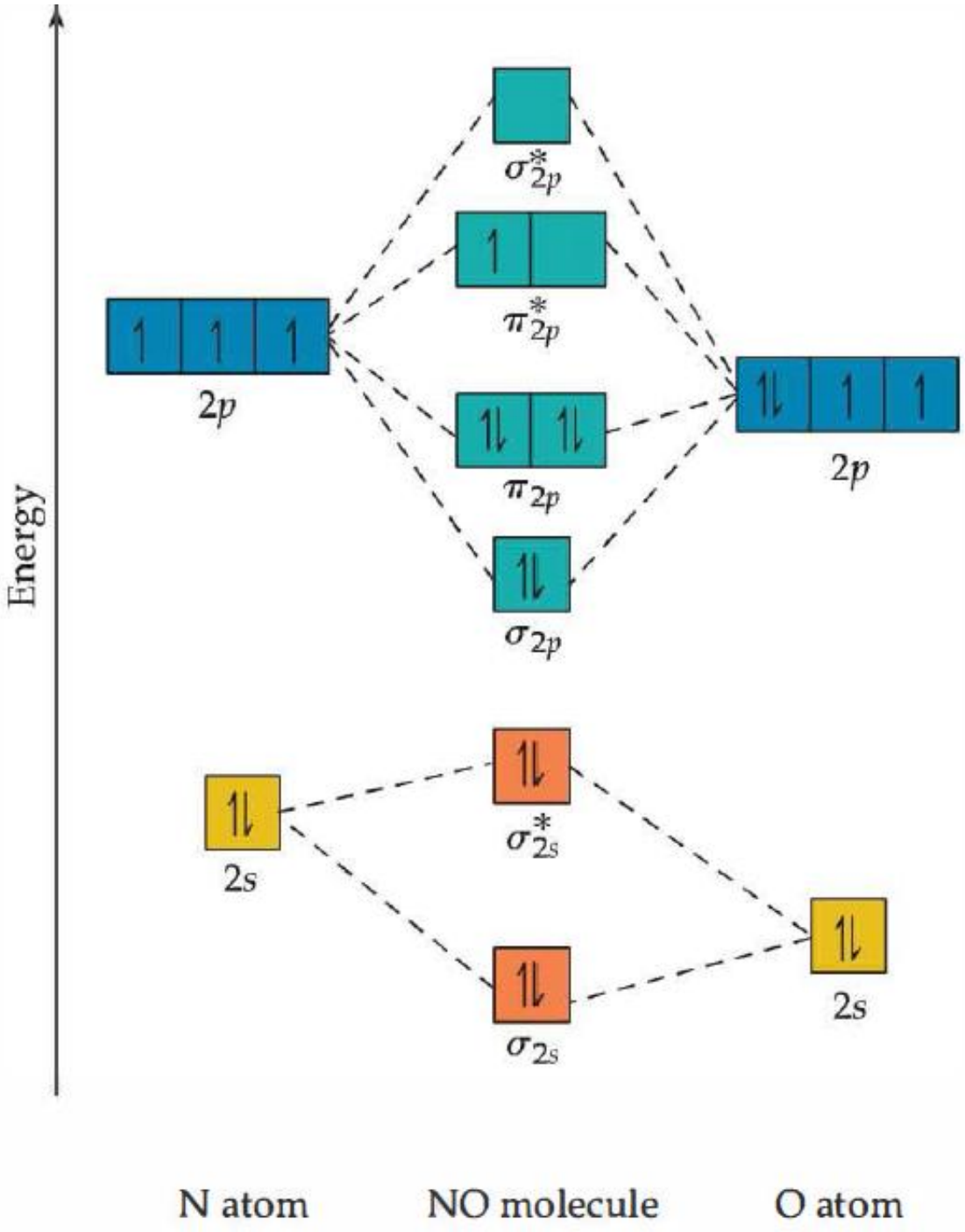
Second-row Diatomic Molecules



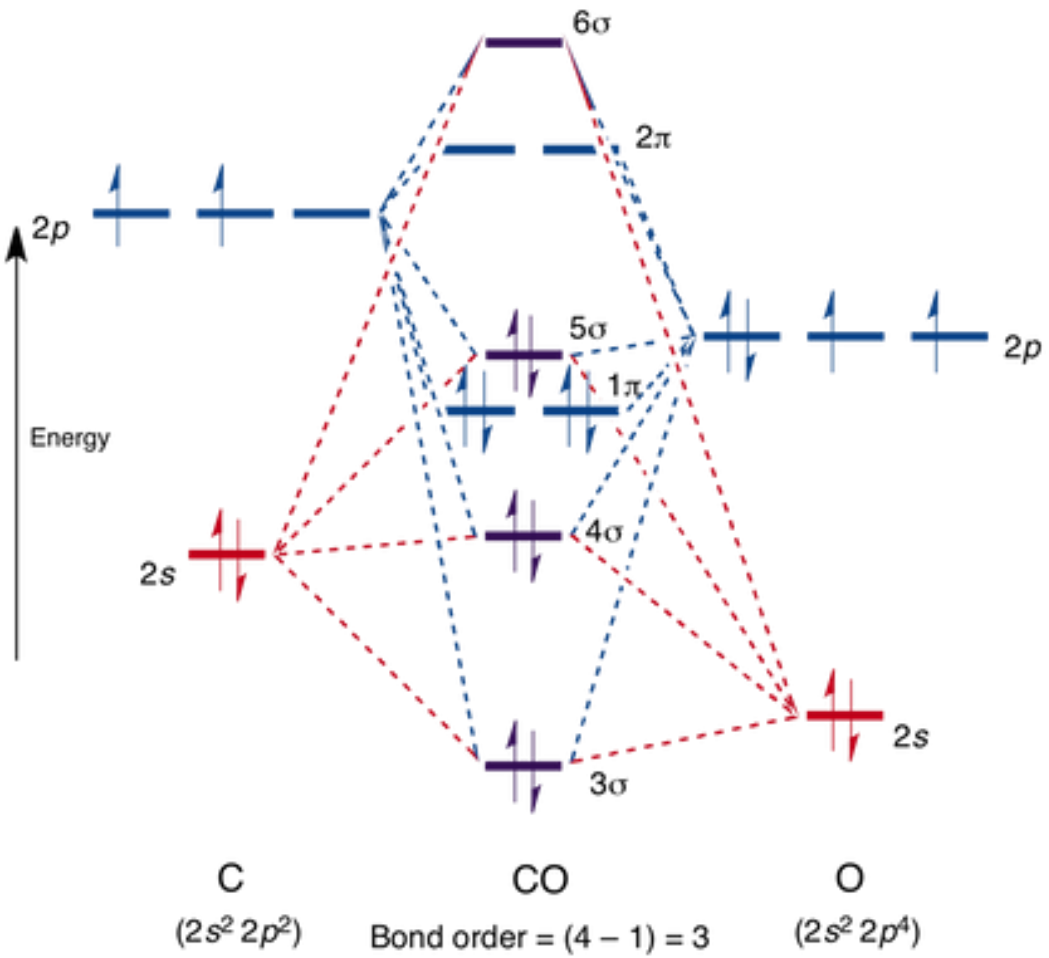
	O_2	F_2	Ne_2
Bond order	2	1	0
Magnetism	Paramagnetic	Diamagnetic	–

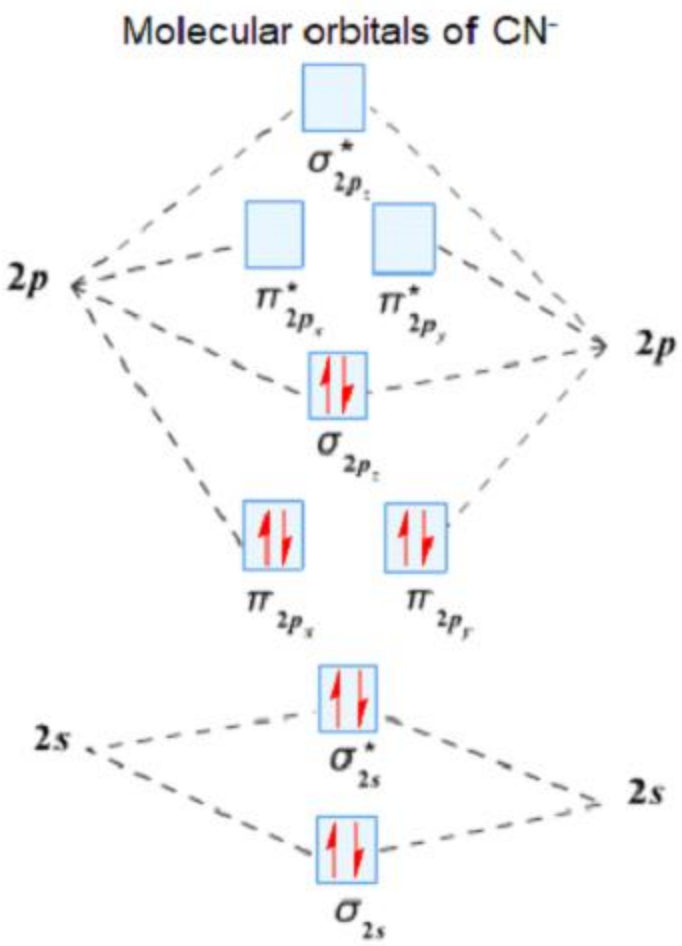
Heteronuclear Diatomic Molecules





CO Molecular orbital diagram

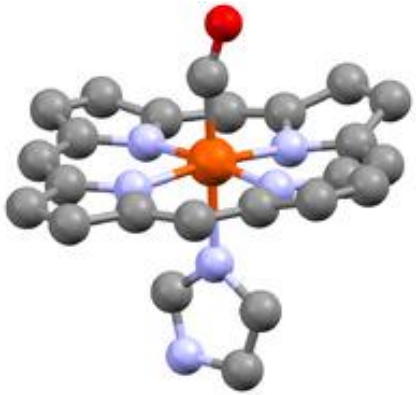



Molecular orbitals of CN-
$\sigma^*_{2p_z}$
2p
$\pi^*_{2p_x}$
$\pi^*_{2p_y}$
2p
σ_{2p_z}
π_{2p_x}
π_{2p_y}
2s
σ^*_{2s}
2s
σ_{2s}

Carboxyhemoglobin or carboxyhaemoglobin (HbCO)

$HbCO + O_2 \rightleftharpoons Hb + CO + O_2 \rightleftharpoons HbO_2 + CO$

HbCO half-life time in blood: 4-6 h
>>> 70- 35 min by administration of O_2
(pure O_2 or with 4-5%CO_2)





GIẢI TRÍ ĐỜI SỐNG XÃ HỘI ĐẸP SỨC KHỎE THẾ GIỚI MUA SẮM

SỰ KIỆN NÓNG #Đại dịch COVID-19 trên toàn cầu #TT Donald Trump mắc Covid-19 #Nuôi con khôn

SỨC KHỎE

Trời lạnh đốt than sưởi ấm khi ngủ, người đàn ông lực lưỡng bất ngờ bị hôn mê

Thứ sáu, 22/12/2017 07:40

Thích 0 Chia sẻ Tweet Bình luận

Đốt than trong phòng từ 10 giờ đêm hôm trước, đến sáng hôm sau không thấy dậy, người nhà vào phòng thì anh H. đã hôn mê, nguy kịch.

Đốt than trong phòng từ 10 giờ đêm hôm trước, đến sáng hôm sau không thấy dậy, người nhà vào phòng thì anh H. đã hôn mê, nguy kịch.

Hôn mê, tổn thương não vì sưởi ấm bằng than

Gần 2 tuần vừa qua, do thời tiết lạnh kéo dài, nên nhiều người đã dùng mọi cách để đối phó, nhất là khi đêm về. Trong đó, có không ít người chọn phương pháp đốt than để sưởi ấm khi ngủ.

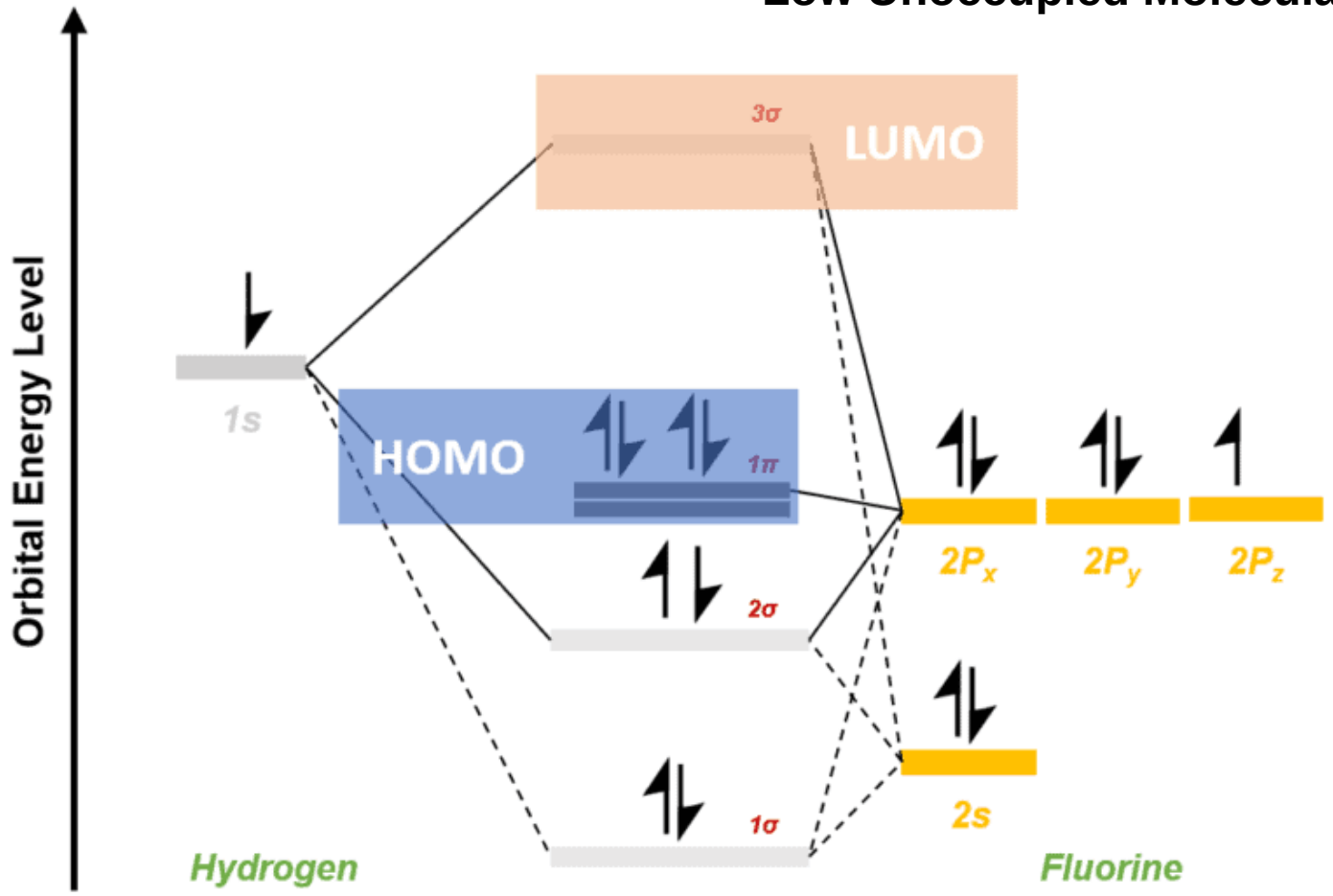
Điển hình như trường hợp của anh N.V.H (42 tuổi, ở Hải Phòng), người nhà anh H cho biết, do nhiệt độ giảm sâu nên ngày 19/12 trước khi đi ngủ (khoảng 10 giờ), anh H có dùng than củi đốt ở trong phòng ngủ nhằm "chống" lạnh.

Đến sáng ngày 20/12, không thấy anh H dậy, người nhà nghĩ do lạnh nên anh H. dậy muộn. Khoảng 10h cùng ngày, vẫn chưa thấy anh H dậy, mọi người cạy cửa vào phòng thì thấy anh đã hôn mê.

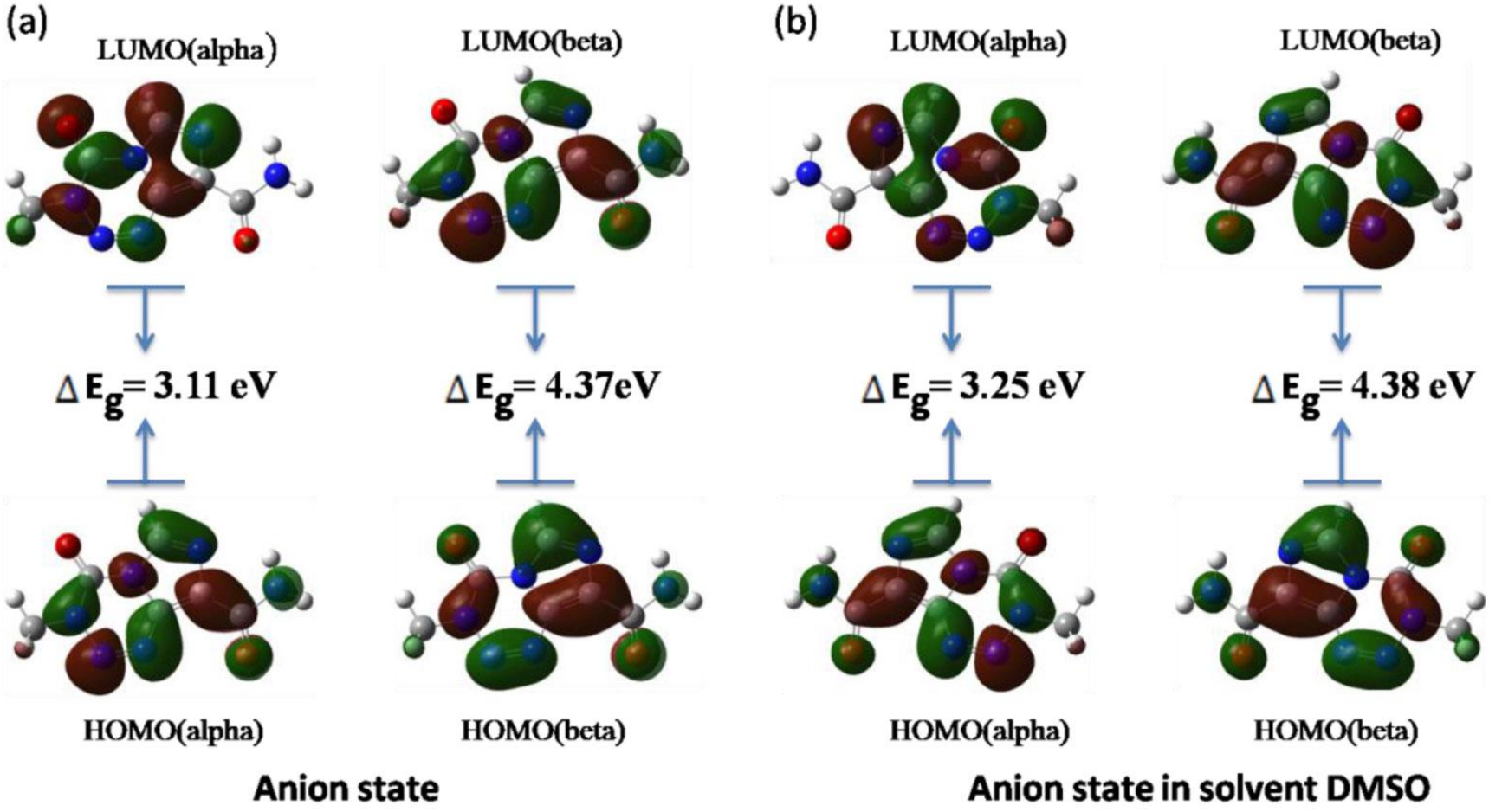


https://tintuconline.com.vn/suc-khoe/troi-lanh-dot-than-suoi-am-khi-ngu-nguoi-dan-ong-luc-luong-bat-ngo-bi-hon-me-n-332245.html

Low Unoccupied Molecular Orbital



High Occupied Molecular Orbital


(a)
LUMO(alpha)
LUMO(beta)
ΔEg= 3.11 eV
ΔEg= 4.37eV
HOMO(alpha)
HOMO(beta)
Anion state
(b)
LUMO(alpha)
LUMO(beta)
ΔEg= 3.25 eV
ΔEg= 4.38 eV
HOMO(alpha)
HOMO(beta)
Anion state in solvent DMSO

Key points

- **Molecular geometry**
  - Valence-shell electron-pair repulsion (VSEPR) model
- **Molecular geometry and polarity**
- **Valence bond theory**
- **Hybridization of atomic orbitals**
  - *s* and *p*
  - *s*, *p,* and *d*
- **Hybridization involving multiple bonds**

Key points

- **Molecular orbital theory**
  - Bonding and antibonding orbitals
  - Sigma (s) molecular orbitals
  - Pi (p) molecular orbitals
  - MO diagrams

General Chemistry I

Chemistry: The central science

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University of Science and Technology of Hanoi

Email: tran-dinh.phong@usth.edu.vn

Lectures 11,12

Bonding theories and **Molecular geometry**